



Full-Integration Implementation Guide

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PAX Technology, Inc.

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Preface

Full-Integration Implementation Guide

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Date	Version	Author	Description
2022/5/26	1.02.00	Keenan Wu	1. Update EDC Type Definition and Transaction Type Definition. 2. Add TransType list for EDC table. 3. Update document format. 4. Add sample code for Initialize, Get Variable, Set Variable, Input Account With EMV, Authorize Card and Get PIN Block. 5. Add screenshot for Show Message, Show Dialog, Show Textbox and ShowMessageCenter. 6. Add AuthorizeCard Flow, CompleteOnlineEMV Flow and GetPinblock Flow. 7. Update InputAccountWithEMV Flow. 8. Removed Q25 from Classic Terminal list.
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Contents

Preface 2

Revision History 1

Contents I

1 Overall Description 1

 1.1 Full-Integration Solution Introduction 1

 1.2 BroadPOS P2PE/TermLink Application Introduction 1

 1.3 BroadPOS P2PE/TermLink Application Features 2

 1.4 TermLink Manager Application (Only used in Android TermLink external calling) 2

2 Transaction Flow 4

 2.1 Manual Flow 4

 2.2 Swipe Flow 5

 2.3 Contact EMV Flow 6

 2.4 Contactless EMV Flow 7

3 Communication Type 8

 3.1 Android BroadPOS P2PE/TermLink Application 8

 3.1.1 Internal Calling Model 8

 3.1.2 External Calling Model (Need Install TermLink Manager Application) 8

 3.2 Classic BroadPOS P2PE/TermLink Application 8

4 Commands 9

 4.1 Initialize 9

 4.1.1 Description 9

 4.1.2 Request 9

 4.1.3 Response 9

 4.1.4 Code Example 9

 4.2 Get Variable 10

4.2.1 Description	10
4.2.2 Request	10
4.2.3 Response.....	10
4.2.4 Code Example	10
4.2.5 Parameter Define	11
4.3 Set Variable	12
4.3.1 Description	12
4.3.2 Request	12
4.3.3 Response.....	13
4.3.4 Code Example	13
4.3.5 Parameter Define	13
4.4 Show Dialog	14
4.4.1 Description	14
4.4.2 Request	14
4.4.3 Response.....	15
4.4.4 Screenshot.....	15
4.5 Get Signature	15
4.5.1 Description	15
4.5.2 Request	15
4.5.3 Response.....	15
4.6 Show Message.....	15
4.6.1 Description	15
4.6.2 Request	16
4.6.3 Response.....	16
4.6.4 Screenshot.....	17
4.7 Clear Message	17
4.7.1 Description	17
4.7.2 Request	17
4.7.3 Response.....	17
4.8 Reset.....	18

4.8.1 Description	18
4.8.2 Request	18
4.8.3 Response.....	18
4.9 Update Resource File	18
4.9.1 Description	18
4.9.2 Request	18
4.9.3 Response.....	19
4.10 Do Signature	19
4.10.1 Description.....	19
4.10.2 Request.....	19
4.10.3 Response	19
4.11 Reboot.....	19
4.11.1 Description.....	19
4.11.2 Request.....	19
4.11.3 Response	19
4.12 Get PIN Block	20
4.12.1 Description.....	20
4.12.2 Request.....	20
4.12.3 Response	20
4.12.4 Code Example	21
4.13 Reset MSR.....	23
4.13.1 Description.....	23
4.13.2 Request.....	23
4.13.3 Response	23
4.14 Input Text	23
4.14.1 Description.....	23
4.14.2 Request.....	23
4.14.3 Response	24
4.15 Check File	24
4.15.1 Description.....	24

4.15.2 Request.....	24
4.15.3 Response.....	24
4.16 Authorize Card	24
4.16.1 Description.....	24
4.16.2 Request.....	24
4.16.3 Response.....	26
4.16.4 Code Example	26
4.17 Complete Online EMV	29
4.17.1 Description.....	29
4.17.2 Request.....	29
4.17.3 Response.....	29
4.18 Remove Card	30
4.18.1 Description.....	30
4.18.2 Request.....	30
4.18.3 Response.....	30
4.19 Get EMV TLV Data	30
4.19.1 Description.....	30
4.19.2 Request.....	30
4.19.3 Response.....	30
4.20 Set EMV TLV Data.....	31
4.20.1 Description.....	31
4.20.2 Request.....	31
4.20.3 Response.....	31
4.21 Input Account With EMV	31
4.21.1 Description.....	31
4.21.2 Request.....	31
4.21.3 Response.....	34
4.21.4 Code Example	36
4.22 Show Textbox	39
4.22.1 Description.....	39

4.22.2 Request.....	40
4.22.3 Response.....	40
4.22.4 Screenshot.....	41
4.23 Card Insert Detection	41
4.23.1 Description.....	41
4.23.2 Request.....	41
4.23.3 Response	42
4.24 Session Key Inject	42
4.24.1 Description.....	42
4.24.2 Request.....	42
4.24.3 Response	42
4.25 MAC Calculation.....	43
4.25.1 Description.....	43
4.25.2 Request.....	43
4.25.3 Response	43
4.26 Get PED Information	44
4.26.1 Description.....	44
4.26.2 Request.....	44
4.26.3 Response	44
4.27 Increase KSN.....	45
4.27.1 Description.....	45
4.27.2 Request.....	45
4.27.3 Response	45
4.28 VAS Set Merchant parameters	45
4.28.1 Description.....	45
4.28.2 Request.....	45
4.28.3 Response	45
4.29 VAS Push Data	46
4.29.1 Description.....	46
4.29.2 Request.....	46

4.29.3 Response.....	46
4.30 Upload FTP.....	46
4.30.1 Description.....	46
4.30.2 Request.....	46
4.30.3 Response.....	46
4.31 ShowMessageCenter.....	47
4.31.1 Description.....	47
4.31.2 Request.....	47
4.31.3 Response.....	47
4.31.4 Screenshot.....	48
4.32 GetFormList	48
4.32.1 Description.....	48
4.32.2 Request.....	48
4.32.3 Response.....	49
4.33 GetVarList	49
4.33.1 Description.....	49
4.33.2 Request.....	49
4.33.3 Response.....	49
4.34 Set VarList.....	49
4.34.1 Description.....	49
4.34.2 Request.....	49
4.34.3 Response.....	49
4.35 RunForm	49
4.35.1 Description.....	49
4.35.2 Request.....	50
4.35.3 Response.....	50
5 Data Define.....	51
5.1 EDC Type Definition	51
5.2 Transaction Type Definition	51
5.3 TransType list for EDC.....	52

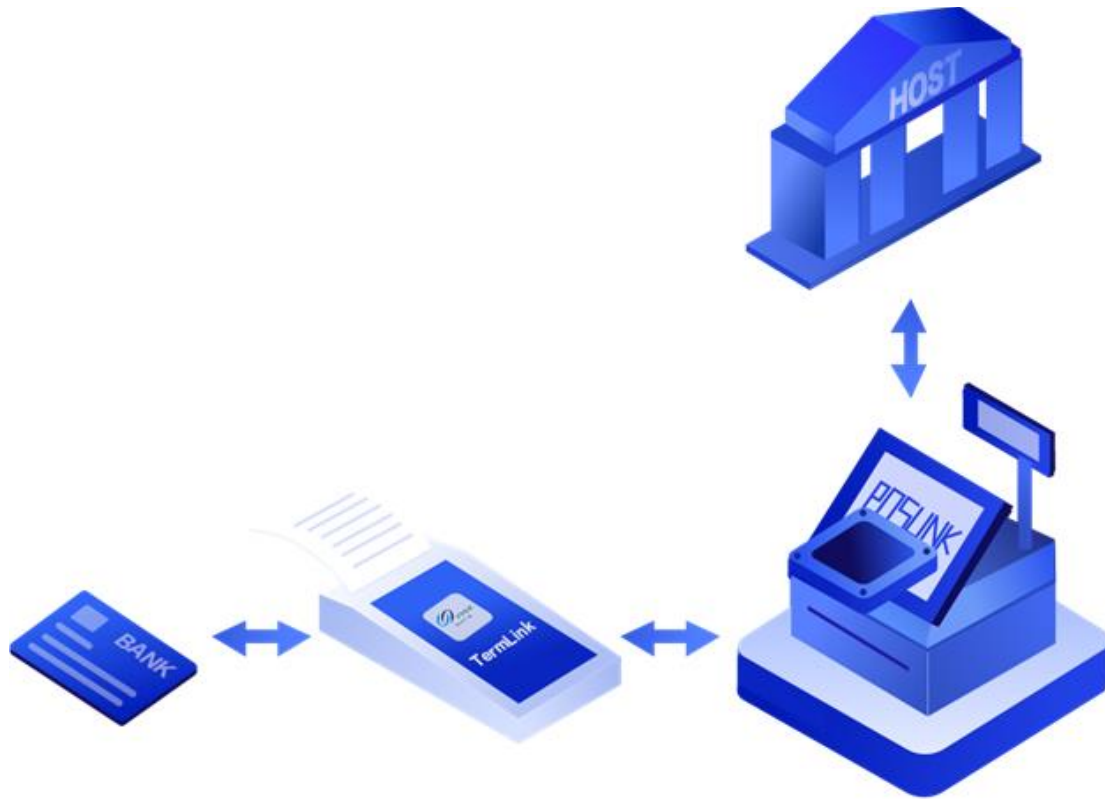
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5.4 EMV Kernel Configuration Type	52
5.5 Encrypted Sensitive TLV Data Format Definition	53
5.5.1 Manual	53
5.5.2 MSR	53
5.5.3 CC/NFC	53
5.6 Key Information Definition	55
5.6.1 DUKPT Key Information Definition	55
5.6.2 Session Key Information Definition	55
5.6.3 Session Key Detail Information Definition	55
5.6.4 AES DUKPT Key Detail Information Definition	55
5.7 VAS Code Definition	55
5.8 Apple Pay VAS Definition	56
5.9 Google Smart Tap Definition	56
5.9.1 Google Smart Tap Capabilities Bitmap Definition	56
5.9.2 Google Service Type Bitmap Definition	57
5.9.3 Google Smart Tap Push Service Definition	57
5.9.4 Google Service Usage Definition	58
5.9.5 Google Service Update Definition	58
5.9.6 Google New Service Definition	58
5.10 Hardware Bitmap Definition	59
6 Response Code	60
6.1 Response code for EMV	60
6.2 Response code for VAS	61
6.3 Response code for PED	61
7 Best Practice (Only for Android App)	63
7.1 Android BroadPOS P2PE/TermLink App Internal Calling	63
7.1.1 Internal Calling Model	63
7.1.2 Transaction Flow With Callback	64
7.1.3 Callback with Single Activity Only	68

7.1.4 Use EventBus	68
7.1.5 API	68
7.2 Android BroadPOS P2PE/TermLink App External Calling (Need Install TermLink Manager Application)	69
7.2.1 External Calling Model.....	69
7.2.2 UI Customization	69
8 Appendix	78
8.1 Classic Terminal EMV Kernel Version	78
8.2 Android Terminal EMV Kernel Version	78
8.3 Offline Data Verification For Contactless.....	79
8.4 Control Character for Text Showing	80
8.5 Terminal Supported By BroadPOS P2PE/TermLink Application	80
8.5.1 Android Terminal	80
8.5.2 Classic Terminal	81

1 Overall Description

1.1 Full-Integration Solution Introduction



Full-Integration

Picture 1 Full-Integration

1.2 BroadPOS P2PE/TermLink Application Introduction

PAX provides an application called BroadPOS P2PE/TermLink for Full-Integration solution. BroadPOS P2PE/TermLink does not store the transactions, nor does it pack messages to any gateway or processor on its own. BroadPOS P2PE/TermLink only reads cards, retrieves track data, PAN, EMV TLV Data, and PIN Blocks through secure entry. The data is returned to the ECR to process online, then the transaction is completed.

Note: The difference between BroadPOS P2PE and TermLink is that BroadPOS P2PE requires that the data must be encrypted (the encryptionFlag must be entered and not zero), but TermLink can set the value of encryptionFlag to 0(data is not encrypted).

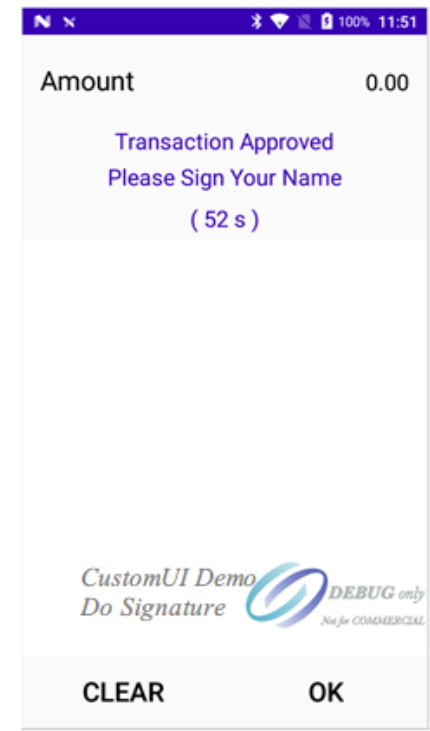
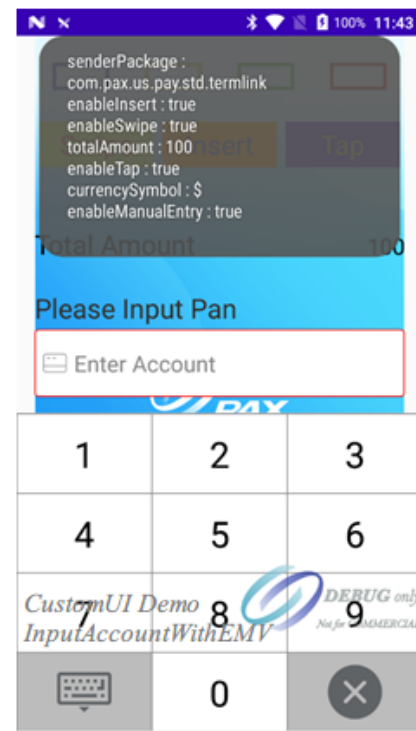
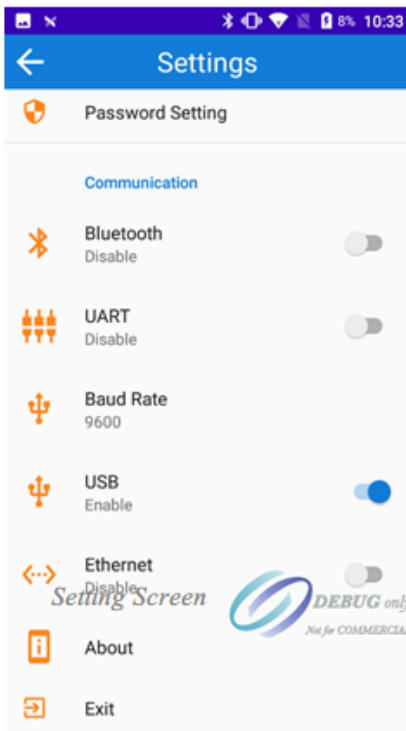
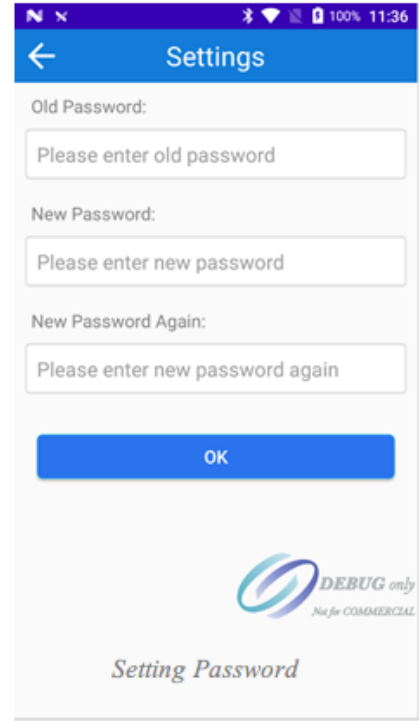
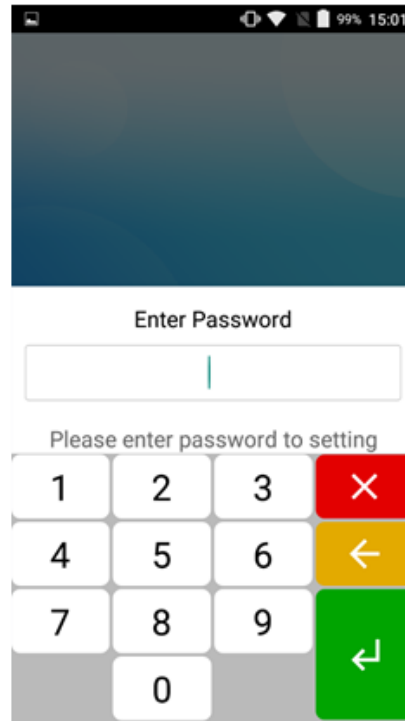
1.3 BroadPOS P2PE/TermLink Application Features

BroadPOS P2PE/TermLink application include Android and Classic version, currently supports InputAccountWithEMV, AuthorizeCard, GetPINBlock, CompleteOnlineEMV and other No-EMV function commands. With these product features, users can get PAN, PIN and other card information from swiping the card, inserting the card, tapping the card or manually typing in the card number, this helps users complete their transaction.

1.4 TermLink Manager Application (Only used in Android TermLink external calling)

For Android BroadPOS P2PE/TermLink, it only supports AIDL communication. It is used in conjunction with POSLink for integration into terminal, as a service. It runs in the background; this limits our application scenarios. To expand more application scenarios, we need to separate POSLink from Android BroadPOS P2PE/TermLink. Since Android BroadPOS P2PE/TermLink is a background service, we need an app that manages different communication types that can be selected by the user based on need.

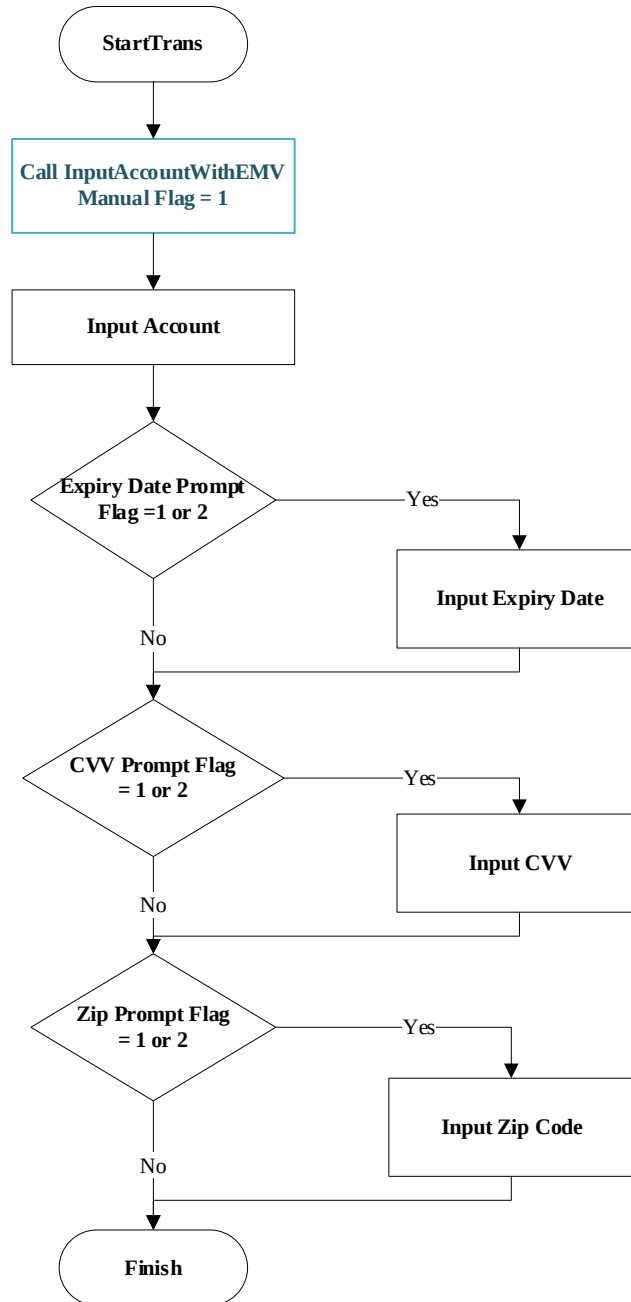
TermLink Manager is an application that evokes BroadPOS P2PE/TermLink service and manages the way app communicates with POS. The terminal will remain idle unless closed with a password. The application includes three parts: Operation, Communication and System. In the operation, we can set the idle display with the current communication type, set the password, and support UI customization. In Communication, we can choose different types of communication, including Bluetooth, UART, USB, and Ethernet (the communication protocol currently supports TCP/SSL/HTTP/HTTPS). In the system, we can get the current BroadPOS P2PE/TermLink service version number and the device SN, and other information.



Picture 2 TermLink Manager Screen

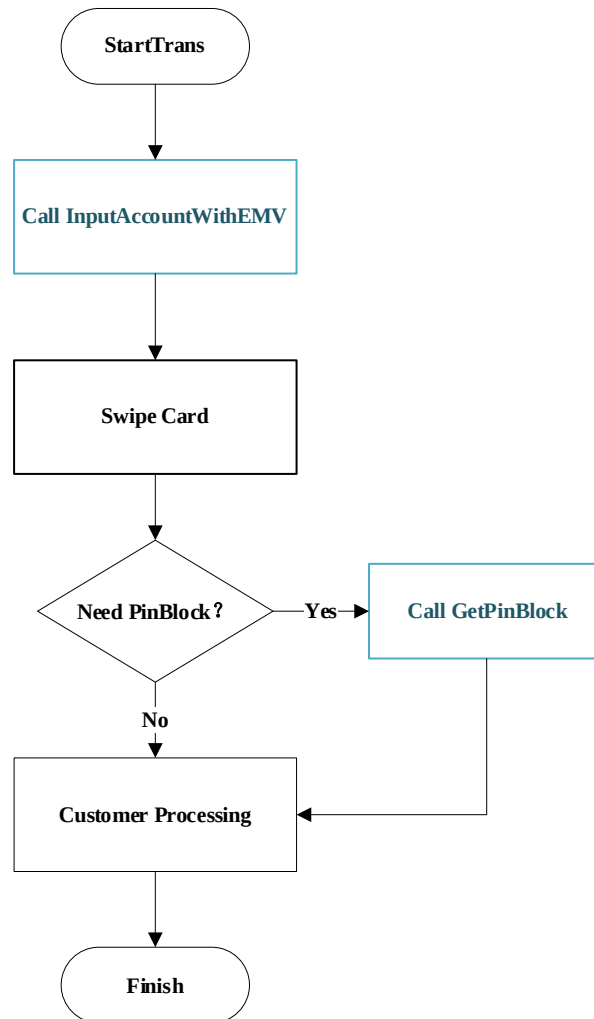
2 Transaction Flow

2.1 Manual Flow



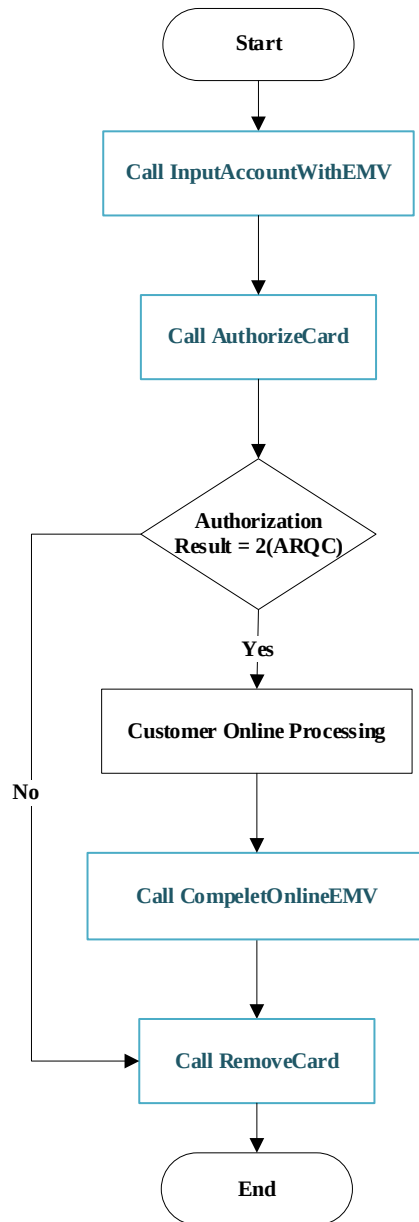
Picture 3 Manual Flow

2.2 Swipe Flow



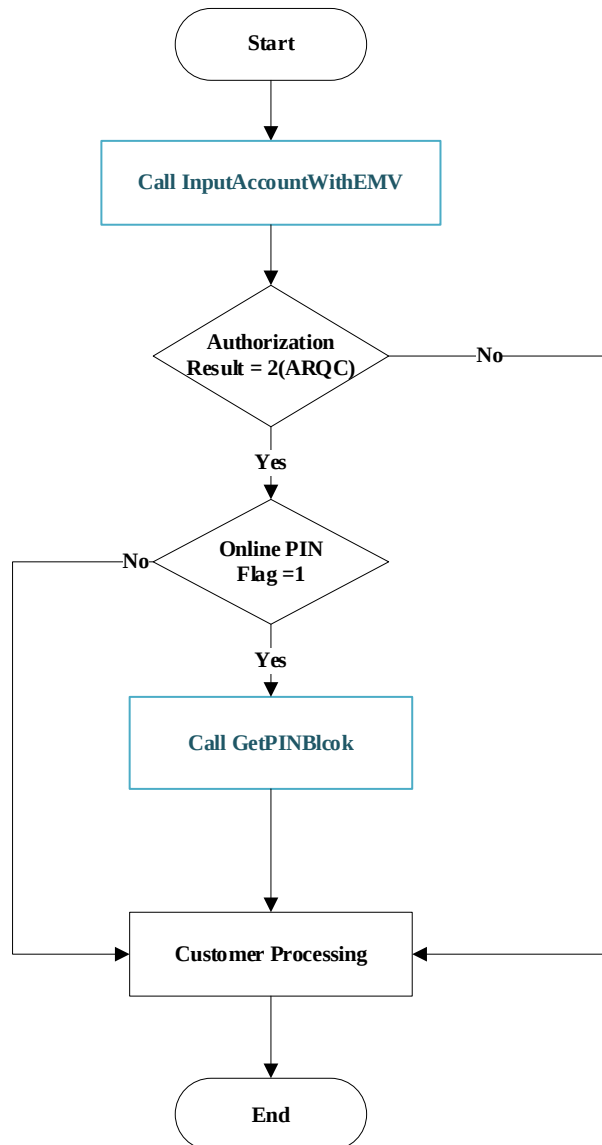
Picture 4 Swipe Card Flow

2.3 Contact EMV Flow



Picture 5 EMV Contact Flow

2.4 Contactless EMV Flow



Picture 6 EMV Contactless Flow

3 Communication Type

3.1 Android BroadPOS P2PE/TermLink Application

3.1.1 Internal Calling Model

A customer can choose internal calling when the POSLink and BroadPOS P2PE/TermLink app are integrated on the same terminal, BroadPOS P2PE/TermLink app only support AIDL communication with POSLink.

3.1.2 External Calling Model (Need Install TermLink Manager Application)

If a customer chooses external calling, TermLink manager application must be installed firstly, customer can set the Communication type through it. BroadPOS P2PE/TermLink application supports Bluetooth/USB/UART/Ethernet(TCP/SSL/HTTP/HTTPS)

3.2 Classic BroadPOS P2PE/TermLink Application

For Classic Terminal and ECR communication, BroadPOS P2PE/TermLink application supports UART/USB/Ethernet/Bluetooth, Ethernet also includes communication types such as TCP/HTTP/HTTPS/SSL.

4 Commands

4.1 Initialize

4.1.1 Description

This command is used to initialize the PAX terminal. The PAX terminal will do internal test/check and initialize the terminal for transactions. It is supported on both classic and Android terminals. Typically, this command is used to get device information.

4.1.2 Request

Command	Field	Description
Init	Null	Null

4.1.3 Response

Command	Field	Description
Init	App Name	App name
	App version	App version
	SN	Terminal SN
	Model Name	Model Name.
	OS Version	OS Version.
	MAC Address	MAC address for LAN module in text.
	LinesPerScreen	Number of lines per screen for ShowMessage command.
	CharsPerLine	Number of characters per line for ShowMessage command.
	WifiMac	Wifi Mac address
	Touchscreen	Indicates whether terminal use a touchscreen Y: Use N: Not use
	Hardware Support Bitmap	Determines what hardware is supported on the device. Please see Hardware Bitmap Definition for details.

4.1.4 Code Example

It should be noted that the following code is not suitable for running in the main thread.

```
1. Init.InitResponse initResponse = Init.init(context,commSetting);
```

Code 1 Initialize Example for Android

```

1. PosLink posLink = new PosLink();
2. ManageRequest manageRequest = new ManageRequest();
3. manageRequest.TransType = manageRequest.ParseTransType("INIT");
4. posLink.ManageRequest = manageRequest;
5. ProcessTransResult result = posLink.ProcessTrans();
6. if (result.Code == ProcessTransResult.ProcessTransResultCode.OK) {
7.     ManageResponse manageResponse = posLink.ManageResponse;
8. }

```

Code 2 Initialize Example for Java

4.2 Get Variable

4.2.1 Description

This command is used to get the control variables and configure parameters in the terminal. It uses the single transaction flow. It is supported on both classic and Android terminals.

4.2.2 Request

Command	Field	Description
GetVariable	EDC Type	EDC Type
	VarName	The name of the variable
	VarName 1	The name of the variable
	VarName 2	The name of the variable
	VarName 3	The name of the variable
	VarName 4	The name of the variable

4.2.3 Response

Command	Field	Description
GetVariable	Result Code	Result Code
	Result txt	If local error message exists, this field is mandatory.
	VarValue	The value of the variable, if the value exists, this field is mandatory.
	VarValue 1	The value of the variable, if the value exists, this field is mandatory.
	VarValue 2	The value of the variable, if the value exists, this field is mandatory.
	VarValue 3	The value of the variable, if the value exists, this field is mandatory.
	VarValue 4	The value of the variable, if the value exists, this field is mandatory.

4.2.4 Code Example

It should be noted that the following code is not suitable for running in the main thread.

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```

1. Variable.GetVariableRequest getVariableRequest = new Variable.GetVariableRequest();
2. getVariableRequest.setVariableName("param_name");
3. Variable.GetVariableResponse getVariableResponse = Variable.getVariable(context, getVariableRequest, commonSetting);

```

Code 3 Get Variable for Android

```

1. PosLink posLink = new PosLink();
2. ManageRequest manageRequest = new ManageRequest();
3. manageRequest.TransType = manageRequest.ParseTransType("GETVAR");
4. manageRequest.VarName = "param_name";
5. posLink.ManageRequest = manageRequest;
6. ProcessTransResult result = posLink.ProcessTrans();
7. if (result.Code == ProcessTransResult.ProcessTransResultCode.OK) {
8.     ManageResponse manageResponse = posLink.ManageResponse;
9. }

```

Code 4 Get Variable for Java

4.2.5 Parameter Define

Parameter Name	Description
contactRetryBeforeFallback	The number of contact chip retries before fallback swipe
e2eIdentityString	The ID String used for Voltage E2EE Encryption
InitializationVector	The initialization vector for DUKPT encryption. When the algorithm is DUKPT DES, the length is 16. When the algorithm is DUKPT AES, the length is 32. If the value is empty, the initialization vector will be filled with 0
e2eeEncryptionMode	The mode used to determine which voltage encryption format terminal should use. Valid values are TEP1X, TEP2X, TEP4.
binWhitelist	The string used for enable and disable BIN Whitelist Y: Enabled N: Disabled
quickChipMode	Enable or disable quick chip mode. Y: Enable N: Disable
quickChipPlaceholderAmount	When quickChipMode is enabled, this will be the default amount to process the ICC if the transaction amount is set to zero.
contactlessReadBeep	Enable or disable the beep when contactless read Y: Enable N: Disable
emvCardRemoveBeepForContactless	Enable or disable EMV Card Remove Beep For Contactless Y: Enable N: Disable
emvCardRemoveBeep	EMV Card Remove Beep 0: No Beep

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	1: One Beep 2: Continuous Beep
swipeTapPriority	The response priority of Swipe or Tap for the terminals which contactless reader near the magnetic stripe reader. In these terminals, such as A920, A920pro, A77, A30, we might get both swipe and tap signal when user swipe. S: Swipe First T: Tap First
KeyPeriod	The injected E2EE key will expire after a certain period of time. This parameter specifies the time span(days). Scope: [1, 365]
autoShowSecuredKeyboard	Auto show secured keyboard when manual input Account and CVV. N: Never Y: Always A: Only manual entry
lineEncryptionMode	The encryption mode for the communication between A80 and SP30. When use BroadPOS P2PE, this parameter must be Y(enable). N: Disabled A: Auto Y: Enabled
EMVContactVersion	Get information about Kernel version of Contact EMV
PaywaveVersion	Get information about Kernel version of Paywave
PaypassVersion	Get information about Kernel version of Paypass
DPASVersion	Get information about Kernel version of D-PAS
ExpresspayVersion	Get information about Kernel version of Expresspay
JSpeedyVersion	Get information about Kernel version of JSpeedy
QPBOCVersion	Get information about Kernel version of QPBOC
FlashVersion	Get information about Kernel version of Flash

4.3 Set Variable

4.3.1 Description

This command is used to set the control variables and configure parameters in the terminal. It is supported on both classic and Android terminals.

4.3.2 Request

Command	Field	Description
SetVariable	EDC Type	EDC Type
	VarName	The name of the variable
	VarValue	The value of the variable
	VarName 1	The name of the variable
	VarValue 1	The value of the variable
	VarName 2	The name of the variable
	VarValue 2	The value of the variable
	VarName 3	The name of the variable
	VarValue 3	The value of the variable
	VarName 4	The name of the variable
	VarValue 4	The value of the variable

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4.3.3 Response

Command	Field	Description
SetVariable	Result Code	Response code
	Response Message	If local error message exists, this field is mandatory.

4.3.4 Code Example

It should be noted that the following code is not suitable for running in the main thread.

```
1. Variable.SetVariableRequest request = new Variable.SetVariableRequest();
2. request.setVariableName("param_name");
3. request.setVariableValue("param_value");
4. Variable.SetVariableResponse response = Variable.setVariable(context,request,commonSetting);
```

Code 5 Set Variable for Android

```
1. PosLink posLink = new PosLink();
2. ManageRequest manageRequest = new ManageRequest();
3. manageRequest.TransType = manageRequest.ParseTransType("SETVAR");
4. manageRequest.VarName = "param_name";
5. manageRequest.VarValue = "param_value";
6. posLink.ManageRequest = manageRequest;
7. ProcessTransResult result = posLink.ProcessTrans();
```

Code 6 Set Variable for Java

4.3.5 Parameter Define

Parameter Name	Description
contactRetryBeforeFallback	The number of contact chip retries before fallback swipe
e2eIdentityString	The ID String used for Voltage E2EE Encryption
InitializationVector	The initialization vector for DUKPT encryption. When the algorithm is DUKPT DES, the length is 16. When the algorithm is DUKPT AES, the length is 32. If the value is empty, the initialization vector will be filled with 0
e2eeEncryptionMode	The mode used to determine which voltage encryption format terminal should use. Valid values are TEP1X, TEP2X, TEP4.
binWhitelist	The string used for enable and disable BIN Whitelist Y: Enabled N: Disabled
quickChipMode	Enable or disable quick chip mode. Y: Enable N: Disable
quickChipPlaceholderAmount	When quickChipMode is enabled, this will be the default amount to process the ICC if the transaction amount is set to zero.
contactlessReadBeep	Enable or disable the beep when contactless read

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	Y: Enable N: Disable
emvCardRemoveBeepForContactless	Enable or disable EMV Card Remove Beep For Contactless Y: Enable N: Disable
emvCardRemoveBeep	EMV Card Remove Beep 0: No Beep 1: One Beep 2: Continuous Beep
swipeTapPriority	The response priority of Swipe or Tap for the terminals which contactless reader near the magnetic stripe reader. In these terminals, such as A920, A920pro, A77, A30, we might get both swipe and tap signal when user swipe. S: Swipe First T: Tap First
KeyPeriod	The injected E2EE key will expire after a certain period of time. This parameter specifies the time span(days). Scope: [1, 365]
autoShowSecuredKeyboard	Auto show secured keyboard when manual input Account and CVV. N: Never Y: Always A: Only manual entry
lineEncryptionMode	The encryption mode for the communication between A80 and SP30. When use BroadPOS P2PE, this parameter must be Y(enable). N: Disabled A: Auto Y: Enabled

4.4 Show Dialog

4.4.1 Description

Shows screen with title and buttons or menu items. It is supported on both classic and Android terminals.

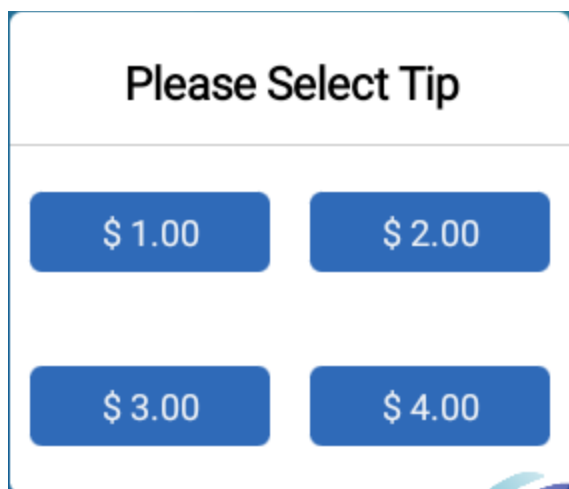
4.4.2 Request

Command	Field	Description
ShowDialog	Title	Title
	Button name 1	The name of button1 on terminal
	Button name 2	The name of button2 on terminal
	Button name3	The name of button3 on terminal
	Button name 4	The name of button4 on terminal
	Timeout	Timeout in 100ms for waiting text input. Valid value should be [0, 9999]. The value of "" means waiting till user canceled or confirmed. The value of "0" means non-blocking mode, only valid when no button.
	Continuous Screen	'0' default. '1' indicate the terminal not go to idle screen but wait for the next command.

4.4.3 Response

Command	Field	Description
ShowDialog	Result code	Result code
	Result message	If local error message exists, this field is mandatory.
	Button number	The button number of selection.

4.4.4 Screenshot



Picture 7 Android screenshot for ShowDialog

4.5 Get Signature

4.5.1 Description

This command is used to get the signature image file from the terminal. It is supported on both classic and Android terminals.

4.5.2 Request

Command	Field	Description
GetSignature	Null	Null

4.5.3 Response

Command	Field	Description
GetSignature	Result code	Result code
	Result message	If local error message exists, this field is mandatory.
	Signature Data	According to the signature data to generate image

4.6 Show Message

4.6.1 Description

Terminal can display several line messages. The newest message can be displayed on the first line or the last line. It is supported on both classic and Android terminals.

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4.6.2 Request

Command	Field	Description
ShowMessage	Title	Title, Please refer to Control Character for Text Showing
	DisplayMessage	The message needs to be display, it can be controlled to be displayed in different area using different font size. Please refer to Control Character for Text Showing
	DisplayMessage2	The message needs to be display, it can be controlled to be displayed in different area using different font size. Please refer to Control Character for Text Showing
	Topdown	Text pushed in top down or bottom up. Y: top down N: bottom up. Default as Y, and the value is only effective at the initial request.
	TaxLine	Tax line to be displayed. Conditional for models with different screen size.
	TotalLine	Total line to be displayed. Conditional for models with different screen size.
	ImageName	Product image name. The image is preloaded on the terminal. Conditional for models with different screen size and terminal capability
	ImageDescription	Product name or description, Conditional for models with different screen size and terminal capability.
	ItemIndex	Use to reference to a line when added.

4.6.3 Response

Command	Field	Description
ShowMessage	Result code	Response code
	Result message	If local error message exists, this field is mandatory.

4.6.4 Screenshot

PAX Market		
Fresh Grade Legs(W)		
price:\$12.00/Kg	500 g	total:\$ 6.00
Spare Rib of Pork		
price:\$15.00/Kg	300 g	total:\$ 4.50
Oero classical 200g		
price:\$3.50	1 pics	total:\$ 3.50
Juice 500ml		
price:\$2.00	1 pics	total:\$ 2.00
Coca-Cola 0 degree 600ml		
price:\$1.10	3 pics	total:\$ 3.30
Total:\$ 19.30		

Picture 8 Android ShowMessage screenshot

4.7 Clear Message

4.7.1 Description

Clear the message buffer on terminal: if terminal receives this command, and then receives the show message command, the terminal only displays the message getting from the show message command. It is supported on both classic and Android terminals.

4.7.2 Request

Command	Field	Description
ClearMessage	Null	Null

4.7.3 Response

Command	Field	Description
---------	-------	-------------

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ClearMessage	Result code	Response code
	Result message	If local error message exists, this field is mandatory.

4.8 Reset

4.8.1 Description

“Reset” command resets the terminal screen to idle state. It is supported on both classic and Android terminals.

4.8.2 Request

Command	Field	Description
Reset	Null	Null

4.8.3 Response

Command	Field	Description
Reset	Result code	Result code
	Result message	If local error message exists, this field is mandatory.

4.9 Update Resource File

4.9.1 Description

This command is used to update resource file like image and EMV parameter file in the terminal. It is supported on both classic and Android terminals.

Note: Resources that need to be updated must be placed in a ZIP file, and “ResourcePath” refers to the path of a ZIP file.

4.9.2 Request

Command	Field	Description
UpdateResource	ResourcePath	The path of resource
	FileType	0: Resource file(Default). A zip package which contains multiple files. Terminal will replace the files according the file names. 1: Offline RKI key file. A zip package which contains the file which used to inject offline RKI key. 2: Installation package file. The file downloaded from PAXSTORE. The file name is like “TermLink-Q25L_V1.02.00E.zip”. Please check you download correct installation package, otherwise it may corrupt the terminal. In Android terminals, APP and OTA upgrade packages can only support upgrade installation, not downgrade installation.

4.9.3 Response

Command	Field	Description
UpdateResource	Result code	Result code
	Result message	If local error message exists, this field is mandatory.

4.10 Do Signature

4.10.1 Description

This command is used to do signature in the terminal. It is supported on both classic and Android terminals.

4.10.2 Request

Command	Field	Description
DoSignature	EDCType	EDC Type
	Upload	1: need upload to host. 0: doesn't need upload to host.
	HRefNum	Which transaction need do signature. This parameter may be used that upload flag is 1.
	Timeout	This is timeout for doing signature. The timeout unit is 100ms. The valid value should be from 150 to 9999. If value is null, it means no timeout and waiting for user cancelation or confirmation.
	Continuous Screen	'0' default. '1' indicate the terminal not go to idle screen but waiting for the next command.

4.10.3 Response

Command	Field	Description
DoSignature	Result code	Result code
	Result message	If local error message exists, this field is mandatory.

4.11 Reboot

4.11.1 Description

Reboot the terminal. It is supported on both classic and Android terminals.

4.11.2 Request

Command	Field	Description
Reboot	Null	Null

4.11.3 Response

Command	Field	Description
Reboot	Result code	Result code
	Result message	If local error message exists, this field is mandatory.

4.12 Get PIN Block

4.12.1 Description

This command is used to input PIN and calculate pin block. It is supported on both classic and Android terminals.

4.12.2 Request

Command	Field	Description
GetPINBlock	AccountNumber	The length of the account number must from 13 to 19.
	EncryptionType	1: DUKPT. 2: Master Session 3: AES DUKPT
	KeySlot	Key slot number, start from 1. Default to 1.
	PinMinLength	The value must be not less 4, and must be not more than pin max length. Default to 4.
	PinMaxLength	The value must be not less pin min length, and must be not more than 12. Default to 12.
	PIN Bypass	The value is as below, default to 0: 0: not allowed 1: allowed.
	PINAlgorithm	The value is as below, default to 0: 0: ISO9564_0 1: ISO9564_1 3: ISO9564_3
	Timeout	Timeout in 100ms for waiting PIN entry. Valid value should be [0, 3000]. The value of "" or 0 means waiting till user canceled or confirmed.
	EDCType	EDC Type
	TransType	Transaction Type
	Title	The title that to be displayed, default as "Amount: \$0.00".
	Pinpad Type	0: Internal Pinpad. (Default) 1: External Pinpad. (A80 only, if use other device will return param error) 2: RNIB Pinpad.
	KSN Flag	Indicate whether to increase KSN, when MAC key type is 1(DUKPT Key) Default KSN flag is 1(Increase) 0: Not increase. 1: Increase.

4.12.3 Response

Command	Field	Description
GetPINBlock	Result code	Result code
	Result message	If local error message exists, this field is mandatory.
	PinBlock	The PIN block value.
	KSN	The KSN value.
	Pinpad Type	0: Internal Pinpad. 1: External Pinpad.

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4.12.4 Code Example

Callback

To achieve the complete transaction process, the user should implement the following Callback to monitor the events in the transaction process.

```
1. GetPINBlock.CurrentStepCallback currentStepCallback =  
2.     new GetPINBlock.CurrentStepCallback() {  
3.         @Override  
4.         public void onSuccess() {  
5.             // Current step input completed  
6.         }  
7.         @Override  
8.         public void onFail(String code, String message) {  
9.             //The currently step failed  
10.        }  
11.    };
```

Code 7 GetPINBlock.CurrentStepCallback

```
1. GetPINBlock.EnterPinCallBack enterPinCallBack = new GetPINBlock.EnterPinCallBack() {  
2.     @Override  
3.     public void onAddedPinCharacter() {  
4.         //When a PIN char is added, this method will be called back,  
5.         // and you can customize the UI display effect for this  
6.     }  
7.     @Override  
8.     public void onClearPin() {  
9.         //When a PIN char is deleted, this method will be called back  
10.    }  
11. };
```

Code 8 GetPINBlock.EnterPinCallBack

If additional callback methods are needed, you can change *EnterPinCallBack* to *EnterPinCallBackExpand*.


```

1. GetPINBlock.GetPINBlockCallback getPINBlockCallback = new GetPINBlock.GetPINBlockCallback() {
2.     @Override
3.     public void onEnterPinStart() {
4.         // Prompt the customer to start input PIN.
5.         // You can monitor some status information by calling handleInputPinStart.
6.         GetPINBlock.getInstance().handleInputPinStart(enterPinCallBack,currentStepCallback);
7.     }
8. };

```

Code 9 GetPINBlockCallback

```

1. BasePOSLinkCallback<GetPINBlock.GetPINBlockResponse> posLinkCallback =
2.     new BasePOSLinkCallback<GetPINBlock.GetPINBlockResponse>() {
3.         @Override
4.         public void onFinish(GetPINBlock.GetPINBlockResponse getPINBlockResponse) {
5.             //Run when GetPINBlock finished.
6.         }
7.     };

```

Code 10 POSLinkCallback for GetPINBlock

Call GetPINBlock

You can call a GetPINBlock transaction by the following methods. It should be noted that the following code is not suitable for running in the main thread.

```

1. GetPINBlock.GetPINBlockRequest getPINBlockRequest = new GetPINBlock.GetPINBlockRequest();
2. getPINBlockRequest.setAccountNumber("account_number");
3. getPINBlockRequest.setEncryptionType("1"); //1-DUKPT DES
4. getPINBlockRequest.setKeySlot("1");
5. getPINBlockRequest.setPinAlgorithm("1");
6. GetPINBlock.getPinBlock(context,getPINBlockRequest,commSetting,posLinkCallback,getPINBlockCallback);

```

Code 11 GetPINBlock for Android

```

1. PosLink posLink = new PosLink();
2. ManageRequest manageRequest = new ManageRequest();
3. manageRequest.TransType = manageRequest.ParseTransType("GETPINBLOCK");
4. manageRequest.PinAlgorithm = "1";
5. manageRequest.EncryptionType = "1";
6. manageRequest.KeySlot = "1";
7. manageRequest.TimeOut = "300";
8. posLink.ManageRequest = manageRequest;
9. posLink.SetCommSetting(CommSetting);
10. ProcessTransResult result = posLink.ProcessTrans();
11. if (result.Code == ProcessTransResult.ProcessTransResultCode.OK) {
12.     ManageResponse manageResponse = posLink.ManageResponse;
13. }

```

Code 12 GetPINBlock for Java

4.13 Reset MSR

4.13.1 Description

Reset terminal magnetic card reader buffer. It is supported on both classic and Android terminals.

4.13.2 Request

Command	Field	Description
ResetMSR	Null	Null

4.13.3 Response

Command	Field	Description
ResetMSR	Result code	Result code
	Result message	If local error message exists, this field is mandatory.

4.14 Input Text

4.14.1 Description

This command is used to get text input. It is supported on both classic and Android terminals.

4.14.2 Request

Command	Field	Description
InputText	Title	The title that to be displayed, default as empty.
	InputType	Input type. Default as 0. 0 for Alpha-Numeric string. 1 for Numeric string.

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		2 for date in MMDDYYYY format. 3 for time in HHMMSS format. 4 for currency in XXXXXX.XX format. 5 for password format. 6 for phone number as "(xxx) xxx-xxxx".
	MINLength	MIN length. Default to 0. Valid value [0, 32].
	MAXLength	MAX length. Default to 32. Valid value [1, 32].
	DefaultValue	Default input value. Valid if "InputType" is 0 or 1.
	Timeout	Timeout in 100ms for waiting text input. Valid value should be [1, 9999]. The value of "" means waiting till user canceled or confirmed.
	ContinuousScreen	'0' default. '1' indicate the terminal not go to idle screen but waiting for the next command.

4.14.3 Response

Command	Field	Description
InputText	Result code	Result code
	Result message	If local error message exists, this field is mandatory.
	Text	Text input by customer. For currency input, 10000 will be returned if the customer input \$100.00 on device.

4.15 Check File

4.15.1 Description

This command is used to check file existence. It is supported on both classic and Android terminals.

4.15.2 Request

Command	Field	Description
CheckFile	FileName	The file name with the suffix.

4.15.3 Response

Command	Field	Description
CheckFile	Result code	Result code
	Result message	If local error message exists, this field is mandatory. "OK" with Response Code "000000" means file existed
	Checksum	The algorithm is MD5.

4.16 Authorize Card

4.16.1 Description

This command is used to authorize contact EMV card. It is supported on both classic and Android terminals.

4.16.2 Request

Command	Field	Description
---------	-------	-------------

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AuthorizeCard	Amount	Amount Information(Not include Cash Back)
	CashBackAmount	The cash back amount, \$\$\$\$\$\$CC, if the field is NULL, and the switch on terminal turn on, terminal will prompt dialog to enter it.
	MerchantDecision	Merchant decision to the current transaction. The default value is 0. 0: default (no merchant decision) 1: force online 2: force decline
	EncryptionType	This field is effective only when ONLINE PIN is required The value is as below: 1: DUKPT. 2: Master Session 3: AES DUKPT
	KeySlot	This field is effective only when ONLINE PIN is required
	PINMinLength	This field is effective only when PIN is required. The value must be not less 4, and must be not more than pin max length. Default to 4.
	PINMaxLength	This field is effective only when PIN is required. The value must be not less pin min length, and must be not more than 12. Default to 12.
	PINBypass	The value is as below, default to 0: 0: not allow PIN Bypass 1: bypass offline PIN automatically 2: bypass online PIN automatically 3: bypass both offline and online PINs automatically. 4: allow user to bypass both offline and online PINs
	PINAlgorithm	The value is as below, default to 0: 0: ISO9564_0 1: ISO9564_1 3: ISO9564_3
	EmvKernelConfigurationSelection	EMV Kernel Configuration Type
	TransactionDate	Transaction Date (use terminal date by default) Format is YYMMDD.
	TransactionTime	Transaction Time (use terminal time by default) format is HHMMSS.
	CurrencyCode	Transaction Currency Code ("840" by default)
	Currency Exponent	Transaction Currency Exponent ("02" by default)
	Tag List	EMV tag list to be returned in EMV TLV data in response message.
	Timeout	Timeout in 100ms for waiting PIN entry. Valid value should be [0, 3000]. The value of "" or 0 means waiting till user canceled or confirmed.
	Continuous Screen	'0' default. '1' indicate the terminal not go to idle screen but waiting for the next command.
	Pinpad Type	0: Internal Pinpad. (default) 1: External Pinpad. (A80 only, if use other device will return param error) 2: RNIB Pinpad.

	KSN Flag	Indicate whether to increase KSN, when MAC key type is 1(DUKPT Key) Default KSN flag is 1(Increase) 0: Not increase. 1: Increase.
--	----------	--

4.16.3 Response

Command	Field	Description
AuthorizeCard	Result code	Result code
	Result message	If local error message exists, this field is mandatory.
	AuthorizationResult	Indicate the transaction result after 1st GAC: 0: transaction offline approved 1: transaction offline declined 2: transaction online authorization needed
	SignatureFlag	Indicate that if signature is needed 0: Signature NOT needed 1: Signature needed
	PINBypassStatus	0: False 1: True
	PIN Block	This field is effective only when the CVM type ONLINE PIN is applied. The pin block value.
	KSN	This field is effective only when the CVM type ONLINE PIN is applied. The KSN value for online PIN encryption.
	EMVTLVData	EMV TLV data will be returned in consecutive TLV format data based on the Tag List filed in the request message.
	CVM	Cardholder Verification Method (CVM) Codes, 0 - Fail CVM processing 1 - Plaintext Offline PIN Verification 2 - Online PIN 3 - Plaintext Offline PIN and Signature 4 - Enciphered Offline PIN Verification 5 - Enciphered Offline PIN Verification and Signature 6 - Signature 7 - No CVM Required 8 - On Device CVM
	Pinpad Type	0: Internal Pinpad. 1: External Pinpad.

4.16.4 Code Example

Callback

To achieve the complete transaction process, the user should implement the following CallBack to monitor the events in the transaction process.

```

1.  AuthorizeCard.EnterPinCallBack enterPinCallBack = new AuthorizeCard.EnterPinCallBack() {
2.      @Override
3.      public void onAddedPinCharacter() {
4.          //When a PIN char is added, this method will be called back,
5.          // and you can customize the UI display effect for this
6.      }
7.      @Override
8.      public void onClearPin() {
9.          //When a PIN char is deleted, this method will be called back
10.     }
11. };

```

Code 13 AuthorizeCard.EnterPINCallback

If additional CallBacks are needed, whether or not the screen is touched, you can change *EnterPinCallBack* to *EnterPinCallBackExpand*.

```

1.  AuthorizeCard.CurrentStepCallback currentStepCallback =
2.      new AuthorizeCard.CurrentStepCallback() {
3.          @Override
4.          public void onSuccess() {
5.              // Current step input completed
6.          }
7.          @Override
8.          public void onFail(String code, String message) {
9.              //The currently step failed
10.         }
11. };

```

Code 14 CurrentStepCallback for AuthorizeCard

```

1.  AuthorizeCard.AuthorizeCallback authorizeCallback = new AuthorizeCard.AuthorizeCallback() {
2.      @Override
3.      public void onEnterPinStart() {
4.          // Prompt the customer to start input PIN.
5.          // You can monitor some status information by calling handleInputPinStart.
6.          AuthorizeCard.getInstance().handleInputPinStart(enterPinCallBack,currentStepCallback);
7.      }
8.  };

```

Code 15 AuthorizeCallback

```

1. BasePOSLinkCallback<AuthorizeCard.AuthorizeResponse> posLinkCallback =
2.     new BasePOSLinkCallback<AuthorizeCard.AuthorizeResponse>() {
3.         @Override
4.         public void onFinish(AuthorizeCard.AuthorizeResponse authorizeResponse) {
5.             //Run when AuthorizeCard finished.
6.         }
7.     };

```

Code 16 POSLinkCallback for AuthorizeCard

Call AuthorizeCard

You can call a AuthorizeCard transaction by the following methods. It should be noted that the following code is not suitable for running in the main thread.

```

1. AuthorizeCard.AuthorizeRequest authorizeRequest = new AuthorizeCard.AuthorizeRequest();
2. authorizeRequest.setAmount("100");
3. authorizeRequest.setPinEncryptionType("1");
4. authorizeRequest.setKeySlot("1");
5. authorizeRequest.setTimeout(300);
6. AuthorizeCard.authorize(context,authorizeRequest,commSetting,posLinkCallback,authorizeCallback);

```

Code 17 AuthorizeCard for Android

```

1. PosLink posLink = new PosLink();
2. ManageRequest manageRequest = new ManageRequest();
3. manageRequest.TransType = manageRequest.ParseTransType("AUTHORIZECARD");
4. manageRequest.Amount = "100";
5. manageRequest.PinAlgorithm = "1";
6. manageRequest.EncryptionType = "1";
7. manageRequest.KeySlot = "1";
8. manageRequest.TimeOut = "300";
9. posLink.ManageRequest = manageRequest;
10. posLink.SetCommSetting(CommSetting);
11. ProcessTransResult result = posLink.ProcessTrans();
12. if (result.Code == ProcessTransResult.ProcessTransResultCode.OK) {
13.     ManageResponse manageResponse = posLink.ManageResponse;
14. }

```

Code 18 Authcard for Java

4.17 Complete Online EMV

4.17.1 Description

This command is used to complete the online authorized EMV transaction. It is supported on both classic and Android terminals.

4.17.2 Request

Command	Field	Description
CompleteOnlineEMV	OnlineAuthorizationResult	0: transaction approved online 1: transaction declined online 2: connect host failed 3: request partial EMV
	ResponseCode	Host authorization response code, tag 8A. i.e. Contact tag 8A: Y1: Offline approved Y3: Unable to go online, offline approved Z1: Offline declined Z3: Unable to go online, offline declined Other: Online approved or declined
	AuthorizationCode	EMV tag 89, Value generated by the authorization authority for an approved transaction.
	IssuerAuthenticationData	If issuer authentication data (tag 91) is returned by host, it shall be set to terminal
	IssuerScript 1	If issuer script (tag 71) is returned by host, it shall be set to terminal
	IssuerScript 2	If issuer script (tag 72) is returned by host, it shall be set to terminal
	Tag List	EMV tag list to be returned in EMV TLV data in response message.
	Continuous Screen	'0' default. '1' indicate the terminal not go to idle screen but waiting for the next command.

4.17.3 Response

Command	Field	Description
CompleteOnlineEMV	Result code	Result code
	Result message	If local error message exists, this field is mandatory.
	AuthorizationResult	Indicate the below transaction result after 2nd GAC: 0: transaction approved 1: transaction declined Indicate the below transaction result for partial EMV only. 2: partial EMV approve 3: partial EMV decline
	EMVTLVData	EMV TLV data will be returned in consecutive TLV format data based on the Tag List filed in the request message.
	IssuerScriptResults	Issuer script results, HEX format.

4.18 Remove Card

4.18.1 Description

This command is used to prompt cardholder to remove card. It is supported on both classic and Android terminals.

4.18.2 Request

Command	Field	Description
RemoveCard	Message 1	Default message is "Please remove card"
	Message 2	Default message is NULL
	Continuous Screen	'0' default. '1' indicate the terminal not go to idle screen but waiting for the next command.
	Pinpad Type	0: Internal Pinpad. (Default) 1: External Pinpad. (A80 only, if use other device will return param error)

4.18.3 Response

Command	Field	Description
RemoveCard	Result code	Result code
	Result message	If local error message exists, this field is mandatory.
	Pinpad Type	0: Internal Pinpad. 1: External Pinpad.

4.19 Get EMV TLV Data

4.19.1 Description

This command is used to get EMV TLV Data from terminal. It is supported on both classic and Android terminals.

4.19.2 Request

Command	Field	Description
GetEMVTLVData	TLV Type	00 for EMV contact 01 for EMV contactless
	Tag List	List of requested tags. All the tags are consecutive without any separators.

4.19.3 Response

Command	Field	Description
GetEMVTLVData	Result code	Result code
	Result message	If local error message exists, this field is mandatory.
	EMVTLVData	If contact EMV card is read successfully, some related EMV data will be returned in consecutive TLV format data

4.20 Set EMV TLV Data

4.20.1 Description

This command is used to set EMV TLV Data to terminal. It is supported on both classic and Android terminals.

4.20.2 Request

Command	Field	Description
SetEMVTLVData	TLVType	00 for EMV contact 01 for EMV contactless
	EMVTLVData	If contact EMV card is read successfully, some related EMV data will be returned in consecutive TLV format data.

4.20.3 Response

Command	Field	Description
SetEMVTLVData	Result Code	Result Code
	Result message	If local error message exists, this field is mandatory
	Tag List	List of tags which have been set successfully. All the tags are consecutive without any separators.

4.21 Input Account With EMV

4.21.1 Description

This command is used to get account input with EMV card support. It is supported on both classic and Android terminals.

4.21.2 Request

Command	Field	Description
InputAccountWithEMV	EDCType	EDC Type
	TransType	Transaction Type
	Amount	Amount Information(Not include Cash Back)
	CashBackAmt	Cash back amount information
	MagneticSwipeEntryFlag	Whether magnetic card swipe is allowed. Default as 0. 0 : not allowed. 1 : Internal Pinpad. 2 : External Pinpad. (A80 only, if use other device will return param error) 3: External Pinpad first. If external pinpad is present, use external pinpad. If not, use internal pinpad. (A80 only, if use other device will return param error)
	ManualEntryFlag	Whether manual entry is allowed. Default as 0. 0 for not allowed. 1 for allowed.
	ContactlessEntryFlag	Whether contactless entry is allowed. Default as 0. 0 : not allowed. 1 : Internal Pinpad.

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		2 : External Pinpad. (A80 only, if use other device will return param error) 3: External Pinpad first. If external pinpad is present, use external pinpad. If not, use internal pinpad. (A80 only, if use other device will return param error)
	ContactEMVEntryFlag	Whether contact EMV entry is allowed. Default as 0. 0 : not allowed. 1 : Internal Pinpad. 2 : External Pinpad. (A80 only, if use other device will return param error) 3: External Pinpad first. If external pinpad is present, use external pinpad. If not, use internal pinpad. (A80 only, if use other device will return param error)
	FallbackSwipeEntryFlag	Whether fallback swipe is allowed after reading EMV card failed. Default as 0. 0 for not allowed. 1 for allowed.
	FallbackInsertEntryFlag	Whether fallback Insert is allowed after reading EMV card failed. Default as 0. 0 for not allowed. 1 for allowed. 2: Prompt: Bypass Not Allowed
	ScannerFlag	Whether laser scanner entry is allowed. Default as 0. 0 for not allowed. 1 for allowed.
	ExpiryDatePrompt	Whether expiry date needs to be prompted during manual entry. Default as 0. 0: not need to prompt. 1: need to prompt. 2: Prompt: Bypass Not Allowed
	CVVPrompt	Whether Zip code need to prompted during manual entry, Default as 0: 0: not need to prompt. 1: need to prompt. 2: Prompt: Bypass Not Allowed
	ZipCodePrompt	Whether Zip code need to prompted during manual entry, Default as 0: 0: Do not prompt. 1: Prompt. 2: Prompt: Bypass Not Allowed
	EncryptionFlag	Data Encryption flag. Default as 0. 0 for not encrypted. 1 for DUKPT Triple DES Key with CBC mode 2 for DUKPT Triple DES Key with ECB mode 3 for DES/TDES method (Depends on the Key length: TDES for 24 bytes key DES for 8/16 bytes key). 4 for Voltage E2EE. 5 for AES CBC DUKPT. 6 for AES ECB DUKPT.

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		The application will convert it from hexadecimal binary array into an ASCII string. For example, the 3 bytes binary array “\x1a\x23\xd4” can be converted into “1a23d4”.
	KeySlot	Data Encryption key slot number, start from 1.
	Padding Char	This is used to indicate what value needs to pad on data string before encryption. 0: padding 0x00; F: padding 0xFF; The default padding char is 0. Only valid when DatEncryption Flag is 1, 2 or 3.
	MIN Account Length	MIN account length. Default to 10. Valid value [1, 32]. Manual Entry only
	MAX Account Length	MAX account length. Default to 19. Valid value [1, 32]. Manual Entry Only
	EMV Kernel Configuration Selection	EMV Kernel Configuration Selection
	Transaction Date	Transaction Date (use terminal date by default) Format is YYMMDD.
	Transaction Time	Transaction Time (use terminal time by default) format is HHMMSS.
	Currency Code	Transaction Currency Code (“840” by default)
	Currency Exponent	Transaction Currency Exponent (“02” by default)
	Merchant Category Code	Merchant Category Code (“0001” by default)
	Transaction CVM Limit	Configure the limit value of CVM for current transaction. (Only for Visa payWave and only for command Input Account with EMV)
	Tag List	For EMV TRANSACTION ONLY: EMV tag list to be returned in EMV TLV data in response message.
	Timeout	Timeout in 100ms for waiting account entry. Valid value should be [200, 9999].
	Continuous Screen	‘0’ default. ‘1’ indicate the terminal not go to idle screen but waiting for the next command.
	Fallback Insert Entry Flag	Whether fallback insert is allowed after reading contactless card failed. Default as 0. 0 for not allowed. 1 for allowed.
	KSN Flag	Indicate whether to increase KSN, when MAC key type is 1(DUKPT Key) Default KSN flag is 1(Increase) 0: Not increase. 1: Increase.
	ReportStatus	Whether to report status back. “1” to report. “0” not to report.

4.21.3 Response

Command	Field	Description
InputAccountWithEMV	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.
	EntryMode	0: Manual 1: Swipe 2: Contactless 3: Laser scanner 4: Chip 5: Chip Fall Back Swipe 6: Front camera 7: Rear camera
	Track1Data	Track1 data or tag 56, will be encrypted if the data encryption is enabled.
	Track2Data	Track2 data or tag 57 (Track 2 Equivalent Data), will be encrypted if the data encryption is enabled.
	Track3Data	Track3 data, will be encrypted if the data encryption is enabled.
	PAN	Account Number or tag 5A (PAN), will be encrypted if the data encryption is enabled.
	MaskedPAN	First 6 and last 4 Example: 5454545****5454
	BarcodeType	Barcode Type '0'- QRCode '1' – 2D-barcode '2' – 3Dbarcode
	BarcodeData	Barcode Data
	KSN	The KSN value for account encryption as DUKPT.
	Encryption Transmission Block	Encryption Transmission Block. Only valid when Encryption Flag as 4(Voltage E2EE).
	Contactless Transaction Path	For CONTACTLESS TRANSACTION ONLY: The first character represents the card type 10 - contactless PAYPASS (MSD) 11 - contactless PAYPASS (MCHIP) 20 - contactless PAYWAVE (MSD) 21 - contactless PAYWAVE (QVSDC) 22 - contactless PAYWAVE (CONTACT) 23 - contactless PAYWAVE (WAVE2) 24 - contactless PAYWAVE (MSD_CVN17) 25 - contactless PAYWAVE (MSD_LEGACY) 26 - contactless PAYWAVE (VSDC) 30 - contactless DPAS (MSD) 31 – contactless DPAS (EMV) 32 – contactless DPAS (ZIP) 40 - contactless EXPRESSPAY (MSD) 41 - contactless EXPRESSPAY (EMV) 50 - contactless JCB (MSD) 51 - contactless JCB (EMV) 52 - contactless JCB (LEGACY) 53 - contactless JCB (WAVE2) 60 - contactless qPBOC (QVSDC)

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		61 - contactless qPBOC (VSDC) 70 - contactless Flash (EMV)
	AuthorizationResult	For CONTACTLESS TRANSACTION ONLY: Indicate the transaction result: 0: transaction offline approved 1: transaction offline declined 2: transaction online authorization needed
	SignatureFlag	For CONTACTLESS TRANSACTION ONLY: Indicate that if signature is needed 0: Signature is NOT needed 1: Signature is needed
	OnlinePINFlag	For CONTACTLESS TRANSACTION ONLY: Indicate that if online PIN is needed 0: Online PIN is NOT needed 1: Online PIN is needed
	EMVTLVData	For EMV TRANSACTIONS ONLY: EMV TLV data will be returned in consecutive TLV format data based on the Tag List filed in the request message.
	Encrypted EMV TLV Data	Encrypted field 40 EMV TLV Data.
	Encrypted Sensitive TLV Data	Encrypted Sensitive data. Please see the definition format of Encrypted Sensitive TLV Data .
	Expiry Date	The expiration date for manual entry, MMYYY.
	Card Holder Name	The card holder name, if the card holder name exists, this field is mandatory.
	Service Code	The service code of the card
	CVV Code	The CVV code, will be encrypted if the data encryption is enabled.
	Zip Code	The zip code
	CVM	Cardholder Verification Method (CVM) Codes, 0 - Fail CVM processing 1 - Plaintext Offline PIN Verification 2 - Online PIN 3 - Plaintext Offline PIN and Signature 4 - Enciphered Offline PIN Verification 5 - Enciphered Offline PIN Verification and Signature 6 - Signature 7 - No CVM Required 8 - On Device CVM
	VASCODE	VAS response code. refer to section VAS Code Definition
	VASDATA	Tokens returned from card. Separated by <RS>. Please see note.
	NDEFDATA	Google Smart Tap response data in format of NDEF, the message will be encoded with Base64.
	Pinpad Type	0: Internal Pinpad 1: External Pinpad
	Luhn Validation Result	If terminal not support, this field is empty. Otherwise, it will use following value. 0: Not performed 1: Passed

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		2: Failed Please note: 1. Not all card support Luhn. So please make sure the card support Luhn before you use this field. If the card not support Luhn, this value is invalid. 2. Terminal will verify Luhn only when entry mode is Manual or Swipe.
--	--	---

4.21.4 Code Example

Callback

To achieve the complete transaction process, the user should implement the following Callback to monitor the events in the transaction process.

```

1. BasePOSLinkCallback<InputAccount.InputAccountResponse> posLinkCallback =
2.     new BasePOSLinkCallback<InputAccount.InputAccountResponse>() {
3.         @Override
4.         public void onFinish(InputAccount.InputAccountResponse inputAccountResponse) {
5.             //Run when InputAccountWithEMV finished.
6.         }
7.     };

```

Code 19 PosLinkCallback for InputAccountWithEMV

```

1. InputAccount.CurrentStepCallback currentStepCallback =
2.     new InputAccount.CurrentStepCallback() {
3.         @Override
4.         public void onSuccess() {
5.             // Current step input completed
6.         }
7.         @Override
8.         public void onFail(String code, String message) {
9.             //The currently entered information does not meet the specifications,
10.            // for example, the length of the CVV is incorrect.
11.        }
12.    };

```

Code 20 CurrentStepCallback for InputAccountWithEMV

```

1. InputAccount.CardEventListener cardEventListener =
2.     new InputAccount.CardEventListener() {
3.         @Override
4.         public void onEvent(String event) {
5.             //Used to receive the status of card processing,
6.             //Such as contactless light, FALLBACK, etc.
7.             //see ${com.pax.poslink.constant.CardEvent}
8.         }
9.     };

```

Code 21 CardEventListener

```

1. InputAccount.InputAreaParams areaParams = new InputAccount.InputAreaParams();
2. areaParams.height = 100;
3. areaParams.width = 500;
4. areaParams.offsetX = 100;
5. areaParams.offsetY = 300;
6. areaParams.hint = "Please Input XXX";
7. InputAccount.getInstance().sendInputArea(areaParams, new InputAccount.EnterSecurityCallback() {
8.     @Override
9.     public void onAddedSecurityCharacter() {
10.         //When a character is added, this method will be called back,
11.         // and you can customize the UI displayeffect for this
12.     }
13.     @Override
14.     public void onDeletedSecurityCharacter() {
15.         //When a character is deleted, this method will be called back
16.     }
17. });

```

Code 22 SendInputArea

```

1. InputAccount.ProcessingMessageUpdateCallback processingMessageUpdateCallback =
2.     new InputAccount.ProcessingMessageUpdateCallback() {
3.         @Override
4.         public void onUpdate(String s, String s1) {
5.             // Modify the content of the prompt here.
6.         }
7.     };

```

Code 23 ProcessingMessageUpdateCallback


```

1. InputAccount.InputAccountCallback inputAccountCallback = new InputAccount.InputAccountCallbackCompat() {
2.     @Override
3.     public void onInputAccountStart() {
4.         // Prompt the customer to start reading the card.
5.         // You can monitor some status information in the EMV process by calling handleInputAccount.
6.         InputAccount.getInstance().handleInputAccount(currentStepCallback, cardEventListener);
7.         // If you need to manually input, you can display a security flow textbox by calling sendInputArea.
8.         // After completing the input, the following method needs to be called:
9.         InputAccount.getInstance().handleInputCardNum(null, currentStepCallback);
10.    }
11.    @Override
12.    public void onEnterExpiryDate() {
13.        // Prompt the user to enter the ExpiryDate manually.
14.        // After completing the input, the following method needs to be called:
15.        InputAccount.getInstance().handleInputExpiry("MMYY", currentStepCallback);
16.    }
17.    @Override
18.    public void onEnterZip() {
19.        // After completing the input, the following method needs to be called:
20.        InputAccount.getInstance().handleInputZip("zip_data", currentStepCallback);
21.    }
22.    @Override
23.    public void onEnterCVV() {
24.        // You can display a security flow textbox by calling sendInputArea.
25.        // After completing the input, the following method needs to be called:
26.        InputAccount.getInstance().handleInputCVV(null, currentStepCallback);
27.    }
28.    @Override
29.    public void onSelectEMVApp(List<String> list) {
30.        // When the AID needs to be manually selected in the EMV process,
31.        // you need to prompt the user to select here
32.        // Once the selection is complete, you need to inform which one was selected.
33.        InputAccount.getInstance().handleSelectEMVApp(list.get(selectIndex), currentStepCallback);
34.    }
35.    @Override
36.    public void onProcessing(String code, String message) {
37.        // Displaysome necessarystatus information here.
38.        // Some content may change, so you need to call the following methods to monitor
39.        // whether the content changes, or end the prompt.
40.        InputAccount.getInstance().handleProcessing(processingMessageUpdateCallback, currentStepCallback);
41.    }
42.    @Override
43.    public void onWarnRemoveCard() {
44.        // Prompt the user to move the card here, and listening remove status bycall this method:
45.        InputAccount.getInstance().handleWarnRemoveCard(currentStepCallback);
46.    }
47. };

```

Code 24 InputAccountCallback

Call InputAccountWithEMV

You can call an InputAccountWithEMV transaction by the following methods. It should be noted that the following code is not suitable for running in the main thread.

```
1. InputAccount.InputAccountRequest request = new InputAccount.InputAccountRequest();
2. request.setAmount("100");
3. request.setContactEMVEntryFlag("1");
4. request.setMagneticSwipeEntryFlag("1");
5. request.setContactlessEntryFlag("1");
6. request.setEdcType(EDCType.CREDIT);
7. request.setTransType(TransType.SALE);
8. request.setTimeout("300");
9. ...
10. InputAccount.inputAccountWithEMV(context,request,commSetting,posLinkCallback,inputAccountCallback);
```

Code 25 InputAccountWithEMV for Android

```
1. PosLink posLink = new PosLink();
2. ManageRequest manageRequest = new ManageRequest();
3. manageRequest.TransType = manageRequest.ParseTransType("INPUTACCOUNTWITHEMV");
4. manageRequest.Amount = "100";
5. manageRequest.Trans = manageRequest.ParseTrans("SALE");
6. manageRequest.EDCType = manageRequest.ParseEDCType("CREDIT");
7. manageRequest.MagneticSwipeEntryFlag = "1";
8. manageRequest.ContactEMVEntryFlag = "1";
9. manageRequest.ContactlessEntryFlag = "1";
10. manageRequest.TimeOut = "300";
11. posLink.ManageRequest = manageRequest;
12. posLink.SetCommSetting(CommSetting);
13. ProcessTransResult result = posLink.ProcessTrans();
14. if (result.Code == ProcessTransResult.ProcessTransResultCode.OK) {
15.     ManageResponse manageResponse = posLink.ManageResponse;
16. }
```

Code 26 InputAccountWithEMV for Java

4.22 Show Textbox

4.22.1 Description

This command is used for showing a text box on terminal. It is supported on both classic and Android terminals.

4.22.2 Request

Command	Field	Description
ShowTextBox	Title	The title on terminal, See CONTROL-CHARACTER-FOR-TEXT-SHOWING for supported special characters.
	Text	The text to be displayed. Limitation depends on terminal screen size and font size, generally as many as 10 pages can be displayed on terminal. See CONTROL-CHARACTER-FOR-TEXT-SHOWING for supported special characters.
	Button1	The name of button1 on terminal
	Buttoncolor1	Please use the hexadecimal to represent the color. For example, ffffff which represent the white color. 00ff00 which means the green color. If this value not to be set, the default value will be used.
	Button2	The name of button2 on terminal
	Buttoncolor2	Please use the hexadecimal to represent the color. For example, ffffff which represent the white color. 00ff00 which means the green color. If this value not to be set, the default value will be used.
	Button3	The name of button3 on terminal
	Button color3	Please use the hexadecimal to represent the color. For example, ffffff which represent the white color. 00ff00 which means the green color. If this value not to be set, the default value will be used.
	Timeout	Timeout in 100ms for waiting text input. Valid value should be [0, 9999]. The value of "" means waiting till user canceled or confirmed. The value of "0" means non-blocking mode, only valid when no button or signature box.
	ButtonKey1	The hard key to be mapped for Button 1
	ButtonKey2	The hard key to be mapped for Button 2
	ButtonKey3	The hard key to be mapped for Button 3
	EnableHardKey	Whether to disable virtual button displaying. 1 – Enable Hard Key only. 0 – Enable both virtual button and hard keys. 0 as Default.
	HardKeylist	Hard keys separated by <SPACE>, like "KEY0 KEY1".
	SignatureBox	Signature Box: 0 – no signature box(default value) 1 – show signature box
	ContinuousScreen	'0' default. '1' indicate the terminal not go to idle screen but waiting for the next command
	BarcodeType	It can be empty or 7 (QRCODE)
	BarcodeData	Barcode data. For QRCODE, the data length scope is [1, 126]

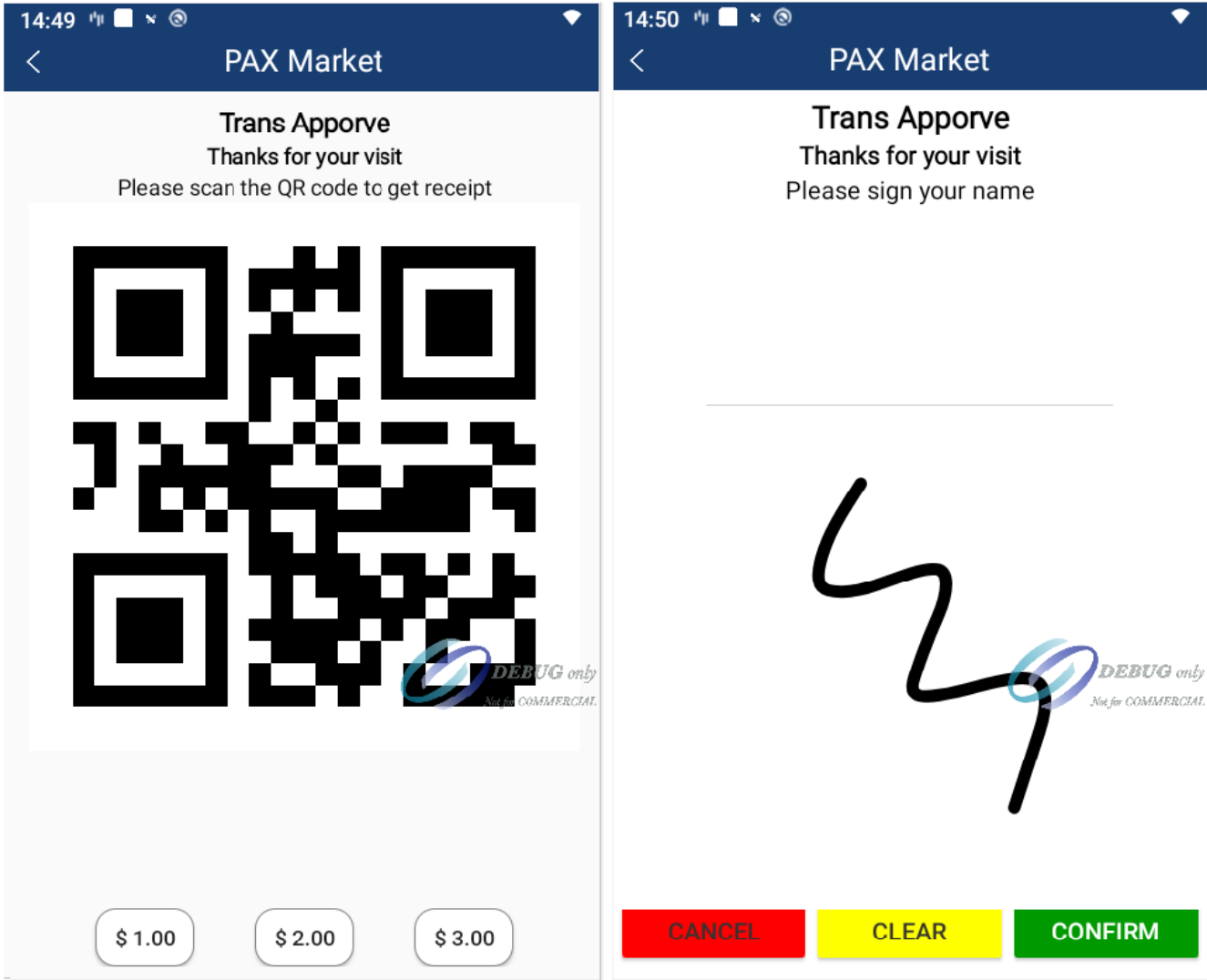
4.22.3 Response

Command	Field	Description
ShowTextBox	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.

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	Button number	The button number of selection.
	SignStatus	Signature status: 1: done the signature.
	Signature Data	Signature data.

4.22.4 Screenshot



Picture 9 Android ShowTextBox screenshot

4.23 Card Insert Detection

4.23.1 Description

It is used to detect whether card inserted or not. It is supported on both classic and Android terminals.

4.23.2 Request

Command	Field	Description
CardInsertDetection	Null	Null

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4.23.3 Response

Command	Field	Description
CardInsertDetection	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.
	CardInsertStatus	0 means no card inserted 1 means has EMV Card inserted 2 means has card inserted, but this is not the EMV card. (Prolin Only)

4.24 Session Key Inject

4.24.1 Description

It is used to inject a session key to a terminal. It is supported on both classic and Android terminals.

4.24.2 Request

Command	Field	Description
SessionKeyInject	SourceKeyType	Support Key Type: TMK(Terminal Master Key) = 1; Note: Now only supports TMK.
	SourceKeyIndex	If source key type = TMK, the source key index should be [1,99].
	DestinationKeyType	TPK(Terminal PIN Key) = 0; TAK(Terminal MAC Key) = 1; TDK(Terminal DES Key) = 2;
	DestinationKeyIndex	The destination key index should be [1, 99].
	DestinationKeyValue	Destination key plaintext or the cipher text, for DES keys, the destination key is the ciphertext which encrypted by the source key in the mode of 3DES ECB. Valid the length of key value is 8, 16, 24. Note: key data must be in ASCII format. For example, the 8 bytes binary array "\x11\x22\x33\x44\x55\x66\x77\x88" can be converted into "1122334455667788".
	CheckMode	0: No KCV check 1: Perform DES/TDES encryption on 8 bytes 0x00, and use first 4 bytes as KCV.
	CheckBuffer	If check mode = 0, PED won't check KCV, this data is no meaning. If check mode = 1, the KCV length should be 4 bytes. Note: the data must be in ASCII format. For example, the 4 bytes binary array "\x11\x22\x33\x44" can be converted into "11223344".

4.24.3 Response

Command	Field	Description
SessionKeyInject	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.

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4.25 MAC Calculation

4.25.1 Description

This command is used for MAC Calculation. It is supported on both classic and Android terminals.

4.25.2 Request

Command	Field	Description
MACCalculation	Input Data	The data used for MAC Calculation. The data can be divided into several parts by vertical bar (' '). Use 'Encryption Bitmap' to indicate which parts is encrypted. Terminal will remove the vertical bars (' ') and decrypt the cipher text, then MAC will be calculated using the decrypted data. The data is represented in hexadecimal. For example: 3031 4a4b 6061.
	Encryption Bitmap	Indicate which parts of input data are encrypted. 0: Plaintext 1: Ciphertext For example: if input data contain 5 parts. 00101 means the third and the fifth part are ciphertext. Default is 0, means this part is plaintext.
	MAC Key Slot	Key Index in PED: range is [1, 99].
	MAC Work Mode	MAC calculation mode: Default is 0. 0: ANSI X9.9. 1: Hypercom Fast Mode. 2: ANSI X9.19.
	Encryption Key Slot	The key slot used to encrypt input data parts. The key can't be DUKPT key.
	Padding Char	This is used to indicate what value had been padded on data when encryption. Terminal will remove the padding chars first when calculate MAC. 0: padding 0x00; F: padding 0xFF; If this field is empty, terminal won't remove the padding chars.
	MAC Key Type	The type of the key which is used to calculate MAC value. Default MAC key type is 0(TAK) 0: TAK 1: DUKPT key
	KSN Flag	Indicate whether to increase KSN, when MAC key type is 1(DUKPT Key) Default KSN flag is 0(Not increase) 0: Not increase. 1: Increase.

4.25.3 Response

Command	Field	Description
MACCalculation	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.

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	Result Data	The MAC result, represented in hexadecimal
	KSN	KSN Value

4.26 Get PED Information

4.26.1 Description

This command is used to get PED information. It is supported on both classic and Android terminals.

4.26.2 Request

Command	Field	Description
GetPEDInformation	Key Type	1 = Master Key 2 = Session Key 3 = DES DUKPT Key 4 = AES DUKPT Key Empty value means all key information. If you want to specify Key Slot, you need to specify key type first
	Key Slot	Slot, Slots or Range of Slot; Use Comma (,) to separate slot numbers and Dash (-) to indicate range Empty value means all slot.

4.26.3 Response

Command	Field	Description
GetPEDInformation	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.
	Available Key Slots Count	Number of key slots remaining available for use, each Key type has its own key slot. Separate by a “,” (comma). The sequence is Master key, Session key, DES DUKPT key. Session key include Terminal Pin Key, Terminal MAC Key, Terminal DES Key. If only specify key type, other values should be empty. For example, if key type in request is 2(session key), this field will be like “,90”. If specify key type and key slot, this field value will be the available key slots count in the specified scope.
	Master Key Info	Terminal Master Key info group. See M/S Key Detail Information Definition .
	Session Key Info	Session Key Info. See Session Key Information Definition .
	DES DUKPT Key Info	DUKPT Key Info group. See DUKPT Key Information Definition .
	AED DUKPT Key Info	AES DUKPT Key Info group. See AES DUKPT Key Information Definition

4.27 Increase KSN

4.27.1 Description

This command is used to get increase KSN. It is supported on both classic and Android terminals.

4.27.2 Request

Command	Field	Description
IncreaseKSN	Key Type	3 = DES DUKPT Key 4 = AES DUKPT Key
	Key Slot	Key slot

4.27.3 Response

Command	Field	Description
IncreaseKSN	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.
	KSN	KSN value after increasing

4.28 VAS Set Merchant parameters

4.28.1 Description

This command is used to configure the parameters of Apple Pay VAS or Google Smart Tap. It is supported on both classic and Android terminals.

4.28.2 Request

Command	Field	Description
VasSetMerchanParameters	Vas Program	'1' – Apple Pay VAS '2' – Google Smart Tap
	VAS Mode	VAS Mode 0 --- VAS_ONLY 1 --- VAS_AND_PAYMENT 2 --- VAS_OR_PAYMENT 3 --- PAYMENT_ONLY
	Apple VAS specific data	please refer to section Apple Pay VAS Definition for detail
	Google VAS specific data	please refer to section Google Smart Tap Definition for detail

4.28.3 Response

Command	Field	Description
VasSetMerchanParameters	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.

4.29 VAS Push Data

4.29.1 Description

This command is used to configure the parameters of Google Smart Tap to push data from terminal to mobile, such as merchant info. It is supported on both classic and Android terminals.

4.29.2 Request

Command	Field	Description
VasPushData	Vas Program	Reserve Must set this field to 2 (Google Smart Tap)
	VAS Mode	VAS Mode 0 --- VAS_ONLY 1 --- VAS_AND_PAYMENT 2 --- VAS_OR_PAYMENT 3 --- PAYMENT_ONLY
	VAS Push Data	Google Smart Tap Push Service. Please refer to Google Smart Tap Push Service Definition for detail.

4.29.3 Response

Command	Field	Description
VasPushData	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.

4.30 Upload FTP

4.30.1 Description

This command is used to tell terminal to upload log files to the FTP server. It is supported on both classic and Android terminals.

4.30.2 Request

Command	Field	Description
UploadFTP	Null	Null

4.30.3 Response

Command	Field	Description
UploadFTP	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.
	SFTP Parameter	Encrypted by CBC TDES. Hexadecimal
	Upload Status	Upload log files status. 0 - OK 1 - COMM ERROR 2 - ABORTED 3 - CONNECT ERROR 4 - UNKNOW ERROR 5 - SFTP COMM ERROR

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4.31 ShowMessageCenter

4.31.1 Description

This command is used to show message center aligned, the default title is "Transaction Approved," and the default message is "Thank You". It is supported on both classic and Android terminals.

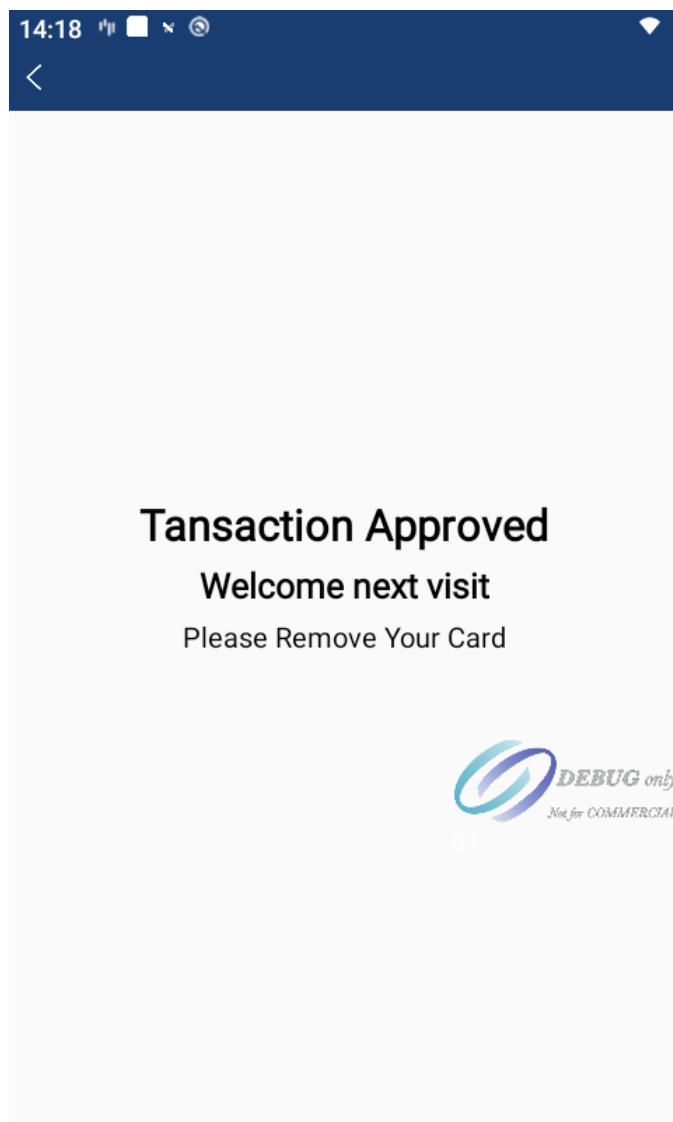
4.31.2 Request

Command	Field	Description
ShowMessageCenter	Title	The title of thank you screen.
	Message1	The message1 of thank you screen. See CONTROL-CHARACTER-FOR-TEXT-SHOWING for supported special characters.
	Message2	The message2 of thank you screen. See CONTROL-CHARACTER-FOR-TEXT-SHOWING for supported special characters.
	TimeOut	Timeout in 100ms. Valid value should be [0, 9999]. The value of "0" means non-blocking mode.
	PinpadType	0: Internal Pinpad. (default). 1: External Pinpad. (A80 only, if use other device will return param error). 3: External Pinpad first. If external pinpad is present, use external pinpad. If not, use internal pinpad. (A80 only, if use other device will return param error).
	Icon Type	0: No icon is displayed. (Default) 1: Approved icon is displayed. 2: Declined icon is displayed.

4.31.3 Response

Command	Field	Description
ShowMessageCenter	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.
	PinpadType	0: Internal Pinpad. (default). 1: External Pinpad. (A80 only, if use other device will return param error). 3: External Pinpad first. If external pinpad is present, use external pinpad. If not, use internal pinpad. (A80 only, if use other device will return param error).

4.31.4 Screenshot



Picture 10 Android ShowMessageCenter screenshot

4.32 GetFormList

4.32.1 Description

This command is designed for Dual Browser TermLink App, it is supported on Q20/IM20 terminal.

4.32.2 Request

Command	Field	Description
GetFormList	FormType	"0" for external forms "1" for internal forms

4.32.3 Response

Command	Field	Description
GetFormList	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.
	FormList	Form List in format: form1,form2,form3,.....

4.33 GetVarList

4.33.1 Description

This command is designed for Dual Browser TermLink App, it is supported on Q20/IM20 terminal.

4.33.2 Request

Command	Field	Description
GetVarList	VarNameList	It uses comma to separate each field Format: varName1, varName2,...

4.33.3 Response

Command	Field	Description
GetVarList	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.
	VarList	Return a list of variable and value pair. JSON format: {“VARNAME1”:“VARVALUE1”,“VARNAME2”:“VARVALUE2”,.....}

4.34 SetVarList

4.34.1 Description

This command is designed for Dual Browser TermLink App, it is supported on Q20/IM20 terminal.

4.34.2 Request

Command	Field	Description
SetVarList	VarList	It uses JSON format:{“VARNAME1”:“VARVALUE1”, “VARNAME2”:“VARVALUE2”}

4.34.3 Response

Command	Field	Description
SetVarList	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.

4.35 RunForm

4.35.1 Description

This command is designed for Dual Browser TermLink App, it is supported on Q20/IM20 terminal.

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4.35.2 Request

Command	Field	Description
RunForm	formName	Form name that want to run
	timeOut	Timeout in 100ms, valid value should be [0, 9999] "" – no timeout, wait till user cancel "0" – non-blocking mode
	Continuous Screen	'0' default. '1' indicate the terminal not go to idle screen but waiting for the next command.

4.35.3 Response

Command	Field	Description
RunForm	Result Code	Result Code
	Result Message	If local error message exists, this field is mandatory.
	Result	Result in format: "RESULT1 RESULT2 RESULT3 RESULT4 RESULT5" maximum 5 results in one API call.

5 Data Define

5.1 EDC Type Definition

EDC Type	Value	Description
ALL	00	All EDC type
CREDIT	01	Credit card
DEBIT	02	Debit card
EBT	03	EBT card(The same rule as Credit)
GIFT	04	Gift card(The same rule as Credit)
LOYALTY	05	Loyalty(The same rule as Credit)
CASH	06	Cash(The same rule as Credit)
CHECK	07	Check(The same rule as Credit)

5.2 Transaction Type Definition

Trans Type	Value	Description
ALL	00	Do not support
SALE	01	Tag 9C: 00(Sale), 09(Sale with cashback)
RETURN	02	Tag 9C: 20(Return)
AUTH	03	Tag 9C: 00(Sale), 09(Sale with cashback)
POST_AUTH	04	Tag 9C: 00(Sale), 09(Sale with cashback)
FORCED	05	Tag 9C: 00(Sale), 09(Sale with cashback)
ADJUST	06	DEFAULT VALUE - Tag 9C: 00(Sale)
WITH DRAWAL	07	DEFAULT VALUE - Tag 9C: 00(Sale)
ACTIVATE	08	DEFAULT VALUE - Tag 9C: 00(Sale)
ISSUE	09	DEFAULT VALUE - Tag 9C: 00(Sale)
ADD	10	DEFAULT VALUE - Tag 9C: 00(Sale)
CASH OUT	11	DEFAULT VALUE - Tag 9C: 00(Sale)
DEACTIVATE	12	DEFAULT VALUE - Tag 9C: 00(Sale)
REPLACE	13	DEFAULT VALUE - Tag 9C: 00(Sale)
MERGE	14	DEFAULT VALUE - Tag 9C: 00(Sale)
REPORT LOST	15	DEFAULT VALUE - Tag 9C: 00(Sale)
VOID	16	DEFAULT VALUE - Tag 9C: 00(Sale)
VOID SALE	17	Tag 9C: 02(Void Sale)
VOID RETURN	18	Tag 9C: 22(Void Return)
VOID AUTH	19	DEFAULT VALUE - Tag 9C: 00(Sale)
VOID POST	20	Tag 9C: 02(Void Sale)
VOID FORCE	21	Tag 9C: 02(Void Sale)
VWITH DRAW	22	DEFAULT VALUE - Tag 9C: 00(Sale)
BALANCE	23	Tag 9C: 30(Balance)
VERIFY	24	DEFAULT VALUE - Tag 9C: 00(Sale)
REACTIVE	25	DEFAULT VALUE - Tag 9C: 00(Sale)
FORCED ISSUE	26	DEFAULT VALUE - Tag 9C: 00(Sale)
FORCED ADD	27	DEFAULT VALUE - Tag 9C: 00(Sale)

UNLOAD	28	DEFAULT VALUE - Tag 9C: 00(Sale)
RENEW	29	DEFAULT VALUE - Tag 9C: 00(Sale)
GET CONVERT DETAIL	30	DEFAULT VALUE - Tag 9C: 00(Sale)
CONVERT	31	DEFAULT VALUE - Tag 9C: 00(Sale)
TOKENIZE	32	DEFAULT VALUE - Tag 9C: 00(Sale)
INCREMENT AUTH	33	DEFAULT VALUE - Tag 9C: 00(Sale)
BALANCE WITH LOCK	34	DEFAULT VALUE - Tag 9C: 00(Sale)
REDEMPTION	35	DEFAULT VALUE - Tag 9C: 00(Sale)
REWARDS	36	DEFAULT VALUE - Tag 9C: 00(Sale)
REENTER	37	DEFAULT VALUE - Tag 9C: 00(Sale)
ADJUSTMENT	38	DEFAULT VALUE - Tag 9C: 00(Sale)
TRANSFER	39	DEFAULT VALUE - Tag 9C: 00(Sale)
REVERSAL	99	DEFAULT VALUE - Tag 9C: 00(Sale)

5.3 TransType list for EDC

EDC Type	Transaction Type
CREDIT	AUTH, SALE, FORCED, RETURN, BALANCE, VOID, VOID SALE, VOID FORCED, VOID RETURN, ADJUST, POST AUTH, VERIFY, TOKENIZE, VOID AUTH, VOID POST, INCREMENT AUTH
DEBIT	SALE, VOID, VOID SALE, BALANCE, RETURN, VOID RETURN, ADJUST, REENTER
EBT	SALE, BALANCE, REFUND, VOID, VOID SALE, VOID RETURN, ADJUST, WITHDRAWAL, REENTER
GIFT	SALE, BALANCE, VOID, UNLOAD, DEACTIVATE, ACTIVATE, RETURN, ADD, ADJUST, CASHOUT, AUTH, POST AUTH, RELOAD, TRANSFER, REPLACE
LOYALTY	BALANCE, VOID, DEACTIVATE, ACTIVATE, ADD, REPLACE, ADJUST, REWARD, RETURN, RENEW, VOID RETURN, TRANSFER, ISSUE, RELOAD
CASH	SALE, RETURN, VOID, VOID SALE, VOID RETURN

5.4 EMV Kernel Configuration Type

EMV Kernel Value	Terminal Type	Online PIN	Offline Enciphered PIN	Offline Plaintext PIN	Signature	No CVM	Comments
01	Attended	√	√	√	√	√	All CVM Supported.
02	Attended	×	√	√	√	×	"No CVM" and "Online PIN" are disabled.
03	Attended	×	√	√	√	√	"Online PIN" is disabled.
04	Attended	√	√	√	√	×	"No CVM" is disabled.
05	Attended	×	×	×	√	×	Only "Signature" is ENABLED.
06	Attended	×	×	×	×	√	Only "No CVM" is ENABLED.
07	Attended	√	√	√	×	√	"Signature" is disabled.
08	Attended	×	×	×	√	√	Only "No CVM" and "Signature" are ENABLED.
09	Unattended	√	√	√	×	√	"Signature" is disabled.

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10	Unattended	√	√	√	×	×	"Signature" and "No CVM" are disabled.
11	Unattended	√	×	×	×	×	Only "Online PIN" is ENABLED.
12	Unattended	×	√	√	×	√	"Signature" and "Online PIN" are disabled.

5.5 Encrypted Sensitive TLV Data Format Definition

5.5.1 Manual

Tag	Data Object	Description	Format	Examples
DFA001	Private	PAN - key entered	cn	PAN Entered: 123456789 TLV: \xDF\xA0\x01\x05\x12\x34\x56\x78\x9F
DFA002	Private	CVV/CID	cn	CVV Entered: 123 TLV: \xDF\xA0\x02\x02\x12\x3F
DFA003	Private	Expiry date (YYMM)	cn	EXPDATE Entered: 1219 TLV: \xDF\xA0\x03\x02\x12\x19

Sample Decrypted Encrypted Sensitive Data:

\xDF\xA0\x01\x05\x12\x34\x56\x78\x9F\xDF\xA0\x02\x02\x12\x3F\xDF\xA0\x03\x02\x12\x19

5.5.2 MSR

Tag	Data Object	Description	Format	Examples
DFA004	Private	Raw magstripe Track2 including start and end sentinel	ascii	Raw Track2: ;4055010000000005=15125025432198712345? TLV: \xDF\xA0\x04\x39;4055010000000005=15125025432198712345?
DFA005 (optional)	Private	Raw magstripe Track1 including start and end sentinel	ascii	

Sample Decrypted Encrypted Sensitive Data:

\xDF\xA0\x04\x39;4055010000000005=15125025432198712345?

5.5.3 CC/NFC

Tag	Data Object	Description	Format	Examples
57	Track 2 Equivalent Data	Contains the data elements of track 2 according to ISO/IEC 7813, excluding start sentinel, end sentinel, and Longitudinal Redundancy Check (LRC)	b	TLV: \x57\x19\x47\x61\x73\x90\x01\x01\x00\x10\xD1\x81\x22\x01\x11\x43\x87\x80\x01\x62\x9F

5A	Application Primary Account Number (PAN)	Valid cardholder account number.	cn	TLV: \x5A\x08\x47\x61\x73\x90\x01\x01\x00\x10
9F6B (optional)	Track 2 Data	Contains the data objects of the Track 2 according to (ISO/IEC 7813), excluding start sentinel, end sentinel and LRC. Optional. MasterCard Only.	Optional b	
56 (optional)	Track1 Data	Track 1 Data contains the data objects of the track 1 according to [ISO/IEC 7813] Structure B, excluding start sentinel, end sentinel and LRC. The Track 1 Data may be present in the file read using the READ RECORD command during a mag-stripe mode transaction.	Optional ans	
9F1F (optional)	Track 1 Discretionary Data	Discretionary part of track 1 according to ISO/IEC 7813	Optional ans	
9F20 (optional)	Track 2 Discretionary Data	Discretionary part of track 2 according to ISO/IEC 7813	Optional cn	

Sample Decrypted Encrypted Sensitive Data:

```
\x57\x19\x47\x61\x73\x90\x01\x01\x00\x10\xD1\x81\x22\x01\x11\x43\x87\x80\x01\x62\x9F\x5A\x08\x47\x61\x73\x90\x01\x01\x00\x10
```

The follow is the illustration for format.

cn - Compressed numeric data elements consist of two numeric digits (having values in the range Hex '0'-'9') per byte. These data elements are left justified and padded with trailing hexadecimal 'F's. Example: The Application Primary Account Number (PAN) is defined as "cn" with a length of up to ten bytes. A value of 1234567890123 may be stored in the Application PAN as Hex '12 34 56 78 90 12 3F FF' with a length of 8.

b - These data elements consist of either unsigned binary numbers or bit combinations that are defined elsewhere in the specification. Binary example: The Application Transaction Counter (ATC) is defined as "b" with a length of two bytes. An ATC value of 19 is stored as Hex '00 13'.

5.6 Key Information Definition

5.6.1 DUKPT Key Information Definition

Field	Description
Key Slot	Key slot with key injected PED
KSN	KSN(Key Serial Number) in terminal PED. corresponding to values returned in Key Slot.
KCV	KCV(Key Check Value) in terminal PED. corresponding to values returned in Key Slot.

5.6.2 Session Key Information Definition

Field	Description
TPK	Terminal Pin Key info group.
TAK	Terminal MAC Key info group
TDK	Terminal Des Key info group.

5.6.3 Session Key Detail Information Definition

Field	Description
Key Slot	Key slot with key injected PED.
KCV	KCV(Key Check Value) in terminal PED, corresponding to values returned in Key Slot.

5.6.4 AES DUKPT Key Detail Information Definition

Field	Description
Key Slot	Key slot with key injected PED
KSN	KSN(Key Serial Number) in terminal PED. corresponding to values returned in Key Slot.
KCV	KCV(Key Check Value) in terminal PED. corresponding to values returned in Key Slot.

5.7 VAS Code Definition

VAS CODE	Description
200	VAS OK.
201	VAS OSE error.
202	VAS get data command error.
203	VAS issue, need to start payment.
204	The app version of mobile is too low.
205	In VAS mode, mobile capabilities: Skip Loyalty Protocol.
206	In dual mode, mobile capabilities: Skip Loyalty Protocol and No Payment Data Forthcoming.
207	In single mode, Mobile Capabilities: Skip Loyalty Protocol and No Payment Data Forthcoming
208	6A83-VAS Data not found (iPhone without the pkpass)
209	6287-VAS Data not active (without fingerprint auth)
210	6984-User Intervention Required on mobile device (a special iOS version will trigger this issue, maybe is without auth, too)

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211	VAS parameter error, The merchant ID is null.
212	VAS get key error, Error when get key from key file
213	VAS Data Decryption Error. Data can't be decrypted by the current key.
255	Unknown error

5.8 Apple Pay VAS Definition

Field	Description
Merchant ID	Max 3 merchant ID connect by '#', Each merchant ID (0, 36]
URL Mode	0 --- Full VAS Mode 1 --- URL mode
URL	Merchant URL, Eg. www.apple.com

5.9 Google Smart Tap Definition

Field	Description
Google Smart Tap CAP	Bitmaps for the various capabilities that the ECR supports. Please see Google Smart Tap Capabilities Bitmap Definition Default value is "00000000000000000000000000000000"
Collect ID	Predefined number specific to the merchant.
Store Local ID	Merchant address code
Terminal ID	Merchant terminal code
Merchant name	Merchant name
Merchant category	Digit merchant code referring to the category of the merchant as defined by the networks.
Service Type	Please refer to Google Service Type Bitmap Definition Default value is "1000000000000000"
Security	0 --- Normal 1 --- Securely 2 --- Pre_signature 3 --- Need not a right auth.
End Tap	0 --- Normal flow (default) 1 --- Stop tap session
OSE TO PPSE	0 --- Normal flow (default) 1--- Stop VAS and do payment directly

5.9.1 Google Smart Tap Capabilities Bitmap Definition

A 32 digits String, 1 for each position means the specified CAP is supported.

Type	Capability
System	Stand-alone
	Semi-integrated
	Unattended
	Online
	Offline
	MMP
	zlib support
	Reserved for future
UI	Printer

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	Printer graphics
	Display
	Images
	Audio
	Animation
	Video
	Reserved for future
Checkout	Support payment
	Support digital receipt
	Support service issuance
	Support OTA POS data
	Reserved for future
	Reserved for future
	Reserved for future
CVM	Reserved for future
	Online PIN
	CD PIN
	Signature
	No CVM
	Device-generated code
	SP-generated code
	ID capture
	Bio-metric

5.9.2 Google Service Type Bitmap Definition

A 16 digits String, 1 for each position means the specified service type is chosen, 0 indicates that the service type is not chosen.

Service Type	Description
All services	This flag must be 0 when specify other flags.
All services except PPSE	This flag must be 0 when specify other flags.
PPSE	
Loyalty	
Offer	
Gift Card 1	
Private Label Card	
Cloud Based Wallet	
Mobile Marketing Platform	
Wallet Customer	

5.9.3 Google Smart Tap Push Service Definition

Field	Description
Security	0 --- Normal 1 --- Securely 2 --- Pre_signature 3 --- Need not a right auth.

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Google Smart Tap CAP	Bitmaps for the various capabilities that the ECR supports. Please see Google Smart Tap Capabilities Bitmap Definition Default value is "00000000000000000000000000000000"
Collect ID	A predefined number specific to the merchant
Service Usage	Please refer to Google Service Usage Definition .
Service Update	Please refer to Google Service Update Definition . Must specify one of
New Service	Please refer to Google New Service Definition
End Tap	0 --- Normal flow (default) 1 --- Stop tap session

Note: Service usage/ Service update/ New service can't be all empty.

5.9.4 Google Service Usage Definition

Field	Description
Usage ID	Service usage ID
State	Service state 0:Unspecified 1:Success 2:Invalid format 3: Invalid value
Title	A short description of valuables. Detailed product information which is not displayed and can be accessed from the URI.
Describe	Intended to be used to provide more context on how the valuable was used.

5.9.5 Google Service Update Definition

Field	Description
Update ID	Service update ID
Update Operation	Provide update operations for application services 0:No operation 1:Remove service object 2:Set balance 3:Add balance 4:Subtract balance 5:Free
Update payload	Details of service updates

5.9.6 Google New Service Definition

Field	Description
Type	New services type 0:Unspecified 1:Valuable 2:Receipt 3:Survey 4:Goods 5:Signup

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Title	Title of the new service
URI	An endpoint where the user may access the new service

5.10 Hardware Bitmap Definition

Position	Property
1	Magstripe
2	EMV Chip
3	EMV Contactless/NFC
4	Camera-Front
5	Laser Scanner
6	Camera-Rear
7	Printer
8	Touchscreen
9	RFU
10	RFU
11	RFU
12	RFU
13	RFU
14	RFU
15	RFU
16	RFU

6 Response Code

6.1 Response code for EMV

Code	Message	Description
000000	OK	The result is OK
100003	1-Chip Malfunction	IC card reset failed
100003	2-Chip Read Error	IC card command failed
100003	3-Chip Blocked	IC card blocked
100003	4-Chip Error	Return code of IC card command is error
100003	5-Chip Blocked	EMV Application is blocked
100003	6-Chip Not Accepted 6-ChipNotAccept	No matching EMV Application in the IC card
100003	7-User Abort	User canceled current operation or transaction
100003	8-User Timeout	User's operation timed out
100003	9-Data Error	IC card data error
100003	10-Chip Not Accepted 10-ChipNotAccept	Transaction is not accepted
100003	11-Chip Reject	Transaction should be declined by devices according to spec.
100003	12-CAPK Expired	CAPK Key Expired
100003	13-PIN Pad Error	There's No PIN Pad or PIN Pad can't be used
100003	14-PIN Pad Error	No PIN Pad or PIN Pad can't be used
100003	15-General Error	Sum of Key Verify is incorrect
100003	16-General Error	Unable to find Appointed Data Element
100003	17-General Error	There is no data in the appointed Data Element
100003	18-General Error	Memory is overflow
100003	19-General Error	No transaction log present
100003	20-General Error	Appointed log does not exist
100003	21-General Error	Appointed Label does not exist in current log record
100003	22-Chip Error	Receive Returned Status Code of 6985 in GPO response.
100003	23-Tap Not Accepted 23-TapNotAccept	Prompt for card entry method other than contactless
100003	24-General Error	EMV File error.
100003	25-Tap Terminated	Must terminate the transaction right now.
100003	26-Tap General Error 26-TapGeneralErr	Contactless failed.
100003	27-Tap Decline	Transaction should be declined by devices according to spec.
100003	28-Tap Not Accepted 28-TapNotAccept	Try another card (DPAS and AMEX).
100003	30-Param Error	Parameter error = EMV_PARAM_ERR
100003	31-Tap General Error 31-TapGeneralErr	International transaction(for VISA AP PayWave Level2 IC card use)
100003	32-Tap General Error 32-TapGeneralErr	Wave2 DDA response TLV format error
100003	33-Tap General Error 33-TapGeneralErr	US card(for VISA AP PayWave L2 IC card use)

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100003	34–Tap General Error 34–TapGeneralErr	Need to use IC card for the transaction(for VISA PayWave IC card use)
100003	35–Tap General Error 35–TapGeneralErr	Need to redo the EMV Application Select.
100003	36–Card Expired	Contactless IC card has expired
100003	37–Tap General Error 37–TapGeneralErr	No application is supported(Select PPSE error)
100003	38–Tap General Error 38–TapGeneralErr	Contactless PBOC
100003	39–Tap CVM Fail	CVM result in decline for AE
100003	40–See Phone	Command response with 6986
100003	41–Tap General Error 41–TapGeneralErr	The last read record command error.
100003	42–Tap General Error 42–TapGeneralErr	Wrong order API calls.
100003	48–Tap General Error 48–TapGeneralErr	Try Tap Again
100003	PIN Pad Timeout	PED Timeout
100003	PIN Pad Failure	PED Failure
100003	105–Declined	Contactless offline declined
100003	108–Too many cards present	Too many cards present.
100003	CLSS ERROR	General contactless error
100003	TERMINAL PARAMETER ERROR	The terminal parameter file is corrupted

6.2 Response code for VAS

Error Code	Error Message	Description
101000	VAS OK, PAYMENT ERROR + ERROR MESSAGE	Apple or Google pay use VAS function. For VAS OK, but payment failed
101001	VAS ERROR, PAYMENT OK	Apple or Google pay use VAS function. For VAS failed, but payment OK
101002	VAS ERROR + (VAS code)	Apple or Google pay use VAS function. For VAS ONLY mode, return error
101003	VAS ERROR, PAYMENT ERROR + ERROR MESSAGE	Apple or Google pay use VAS function. For VAS and Payment both failed
101004	VAS ERROR, PAYMENT NOT PERFORMED	Google Smart Tap failed, and not perform payment.
101005	VAS OK, PAYMENT NOT PERFORMED	Google Smart Tap OK, but not perform payment

6.3 Response code for PED

Error Code	Error Message	Description
100503	PED BUSY	PED is busy
100504	PED INJECTION ERROR	PED injection error
100505	MAC KEY TYPE DISMISMATCH	the specified MAC key type mismatch the specified algorithm

100506	ENCRYPTION KEY TYPE DISMATCH	the encryption key type mismatch the specified algorithm
100507	PED PARAMETER ERROR	PED parameter error
100508	PED GENERAL ERROR	PED general error
100509	PED UNKNOWN ERROR	PED unknown error

7 Best Practice (Only for Android App)

7.1 Android BroadPOS P2PE/TermLink App Internal Calling

7.1.1 Internal Calling Model

Users can put TermLink and POSLink-integrated applications in a terminal to Standalone mode.



Picture 11 Internal Call for Standalone use

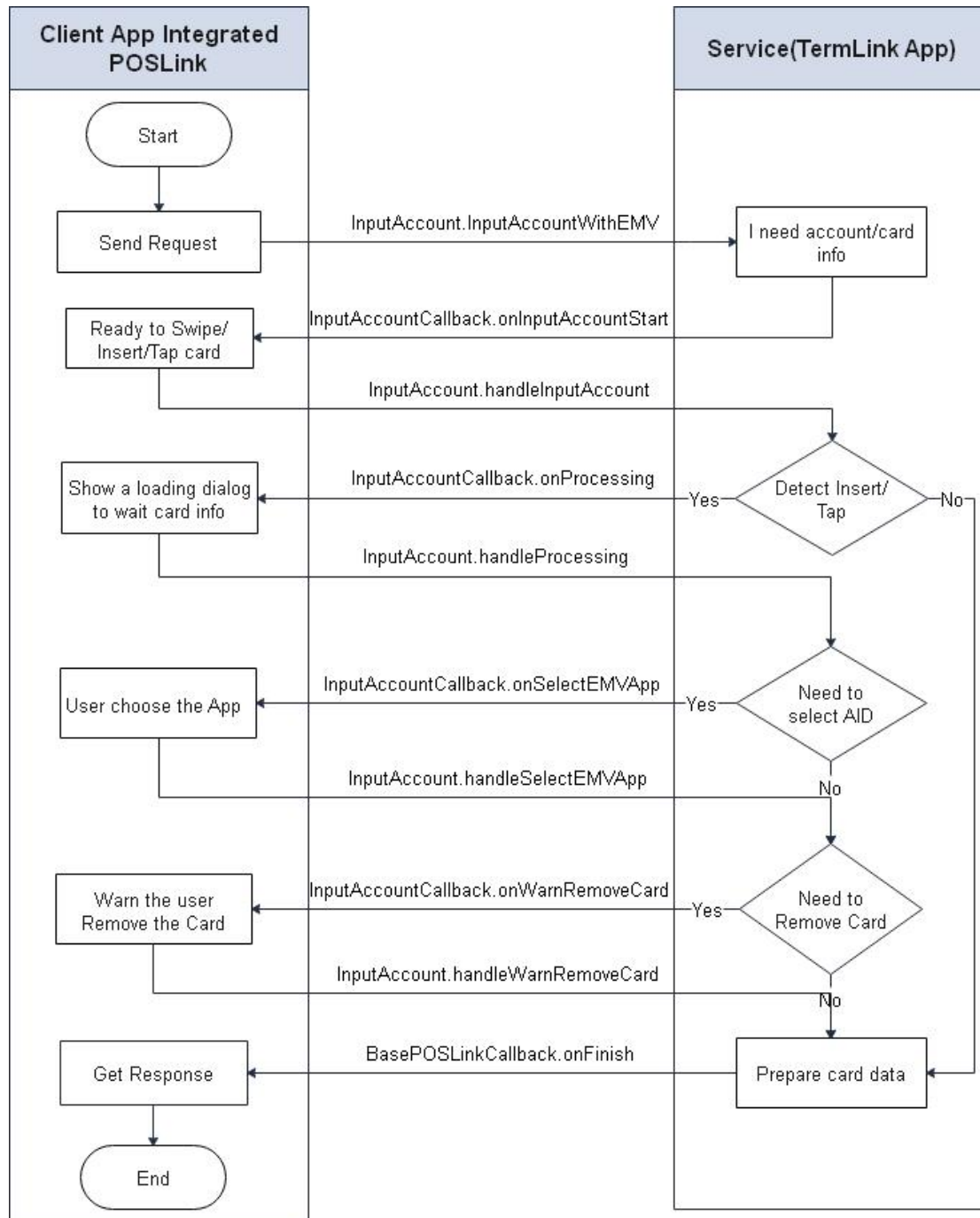
Or communicate with the ECR through a customer-defined protocol by POSLink-integrated application.



Picture 12 Internal Call for ECR

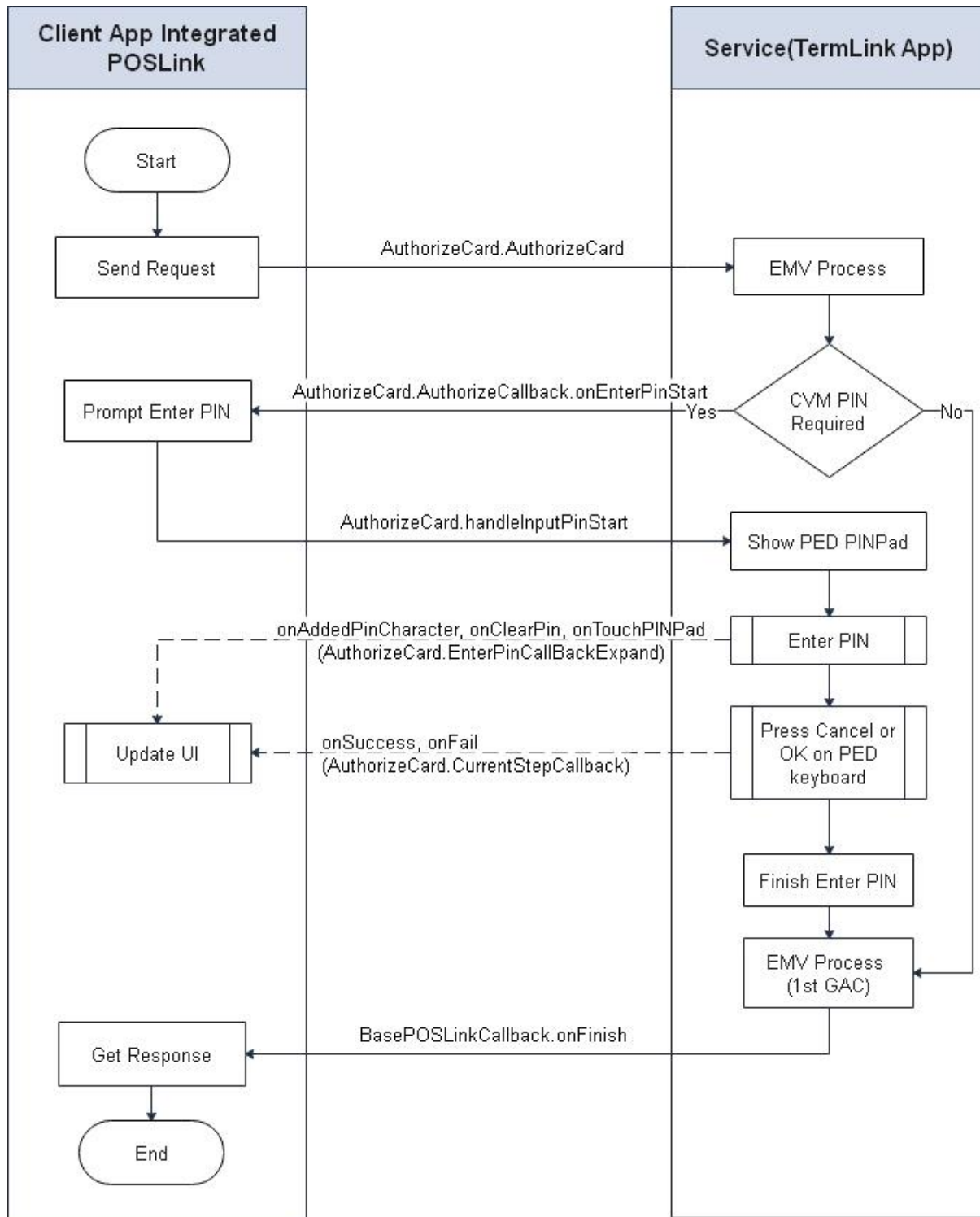
7.1.2 Transaction Flow With Callback

InputAccountWithEMV Flow



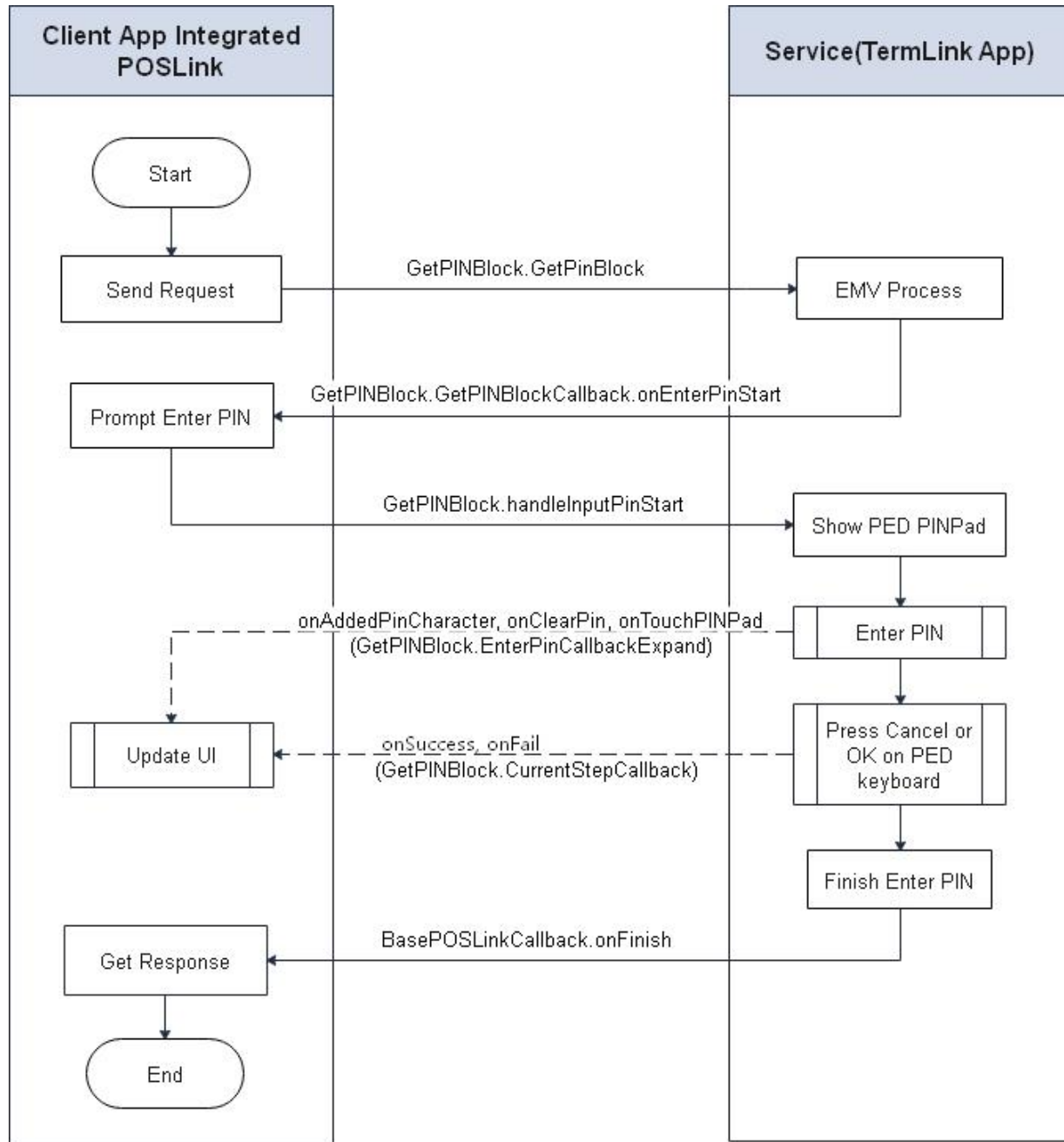
Picture 13 InputAccountWithEMV Callback Flow

AuthorizeCard Flow



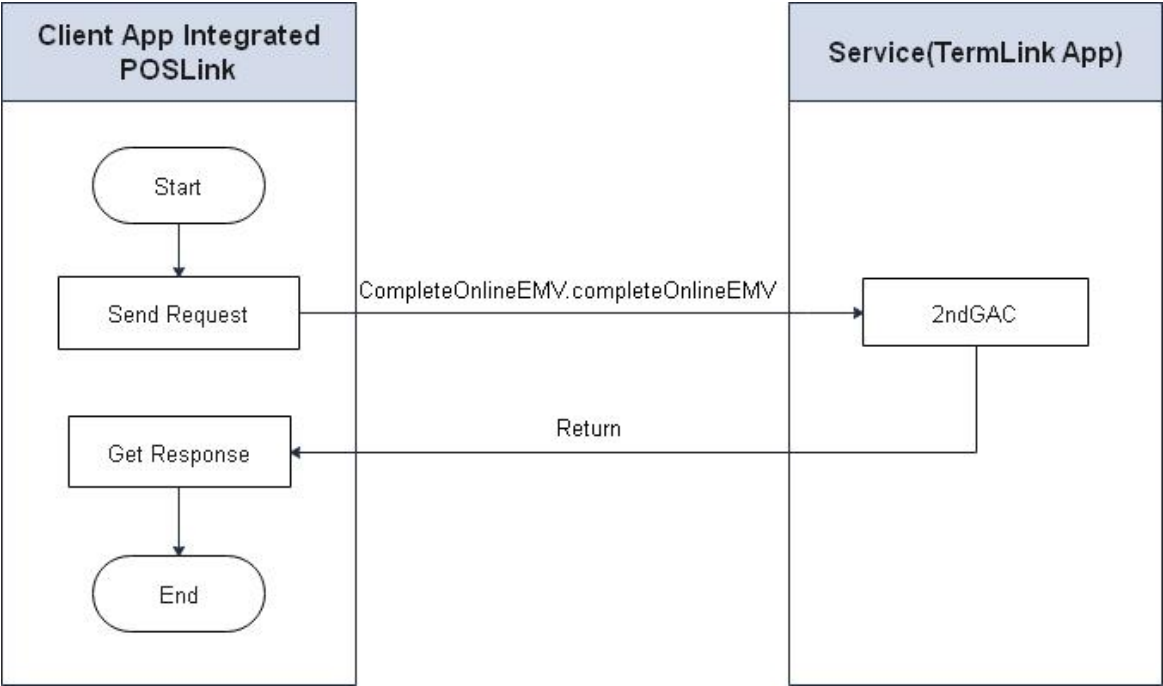
Picture 14 AuthorizeCard Callback Flow

GetPinBlock Flow



Picture 15 GetPinBlock Callback Flow

CompleteOnlineEMV Flow



Picture 16 CompleteOnlineEMV Flow

7.1.3 Callback with Single Activity Only

The structure for InputAccount.InputAccountCallback:

```
1. public interface InputAccountCallback {  
2. ....  
3. onInputAccountStart  
4. onEnterExpiryDate  
5. onSelectEMVApp  
6. onProcessing  
7. onWarnRemoveCard  
8. ....  
9. }
```

Code 27 InputAccount Callback

Each onXXX callback needs the user to display a different view.

As a result, using fragments or dialog to display different views in a Single Activity to help users reduce the complexity of Activity management.

7.1.4 Use EventBus

EventBus is a Java library that simplifies communication between Activities, Fragments, Threads, Services, etc.

Users can get details from <https://github.com/greenrobot/EventBus>

As mentioned above, there are lots of callbacks in InputAccountCallback, so users can separate the callbacks from the user's Activity and use EventBus to communicate between InputAccountCallback and the user's Activity.

7.1.5 API

Please see the POSLink Java document for details.

The main functions are in the package com.pax.poslink.fullintegration. The Class in which the main functions of full-Integration are listed below.

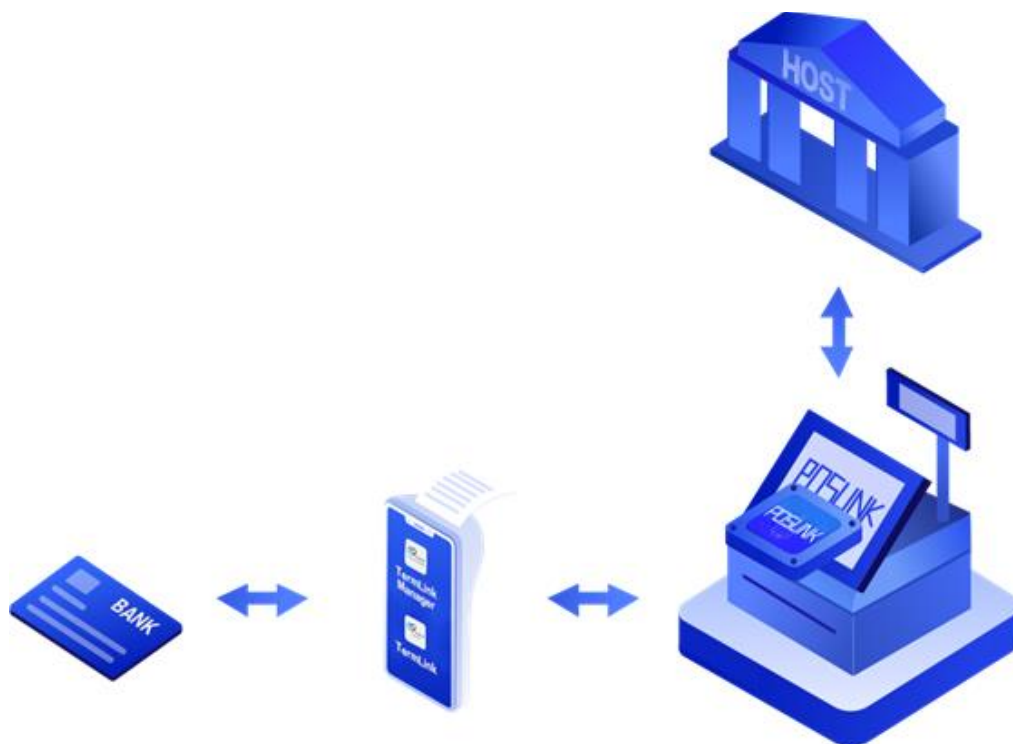
Model Command	Package and class
INIT	com.pax.poslink.fullIntegration.Init
GETVAR	com.pax.poslink.fullIntegration.Variable
SETVAR	com.pax.poslink.fullIntegration.Variable
SHOWDIALOG	com.pax.poslink.formManage.ShowDialog
GETSIGNATURE	com.pax.poslink.formManage.DoSignature
DOSIGNATURE	com.pax.poslink.formManage.DoSignature
UPDATEIMAGE/UPATERESOURCE	com.pax.poslink.fullIntegration.UpdateResource
CHECKFILE	com.pax.poslink.fullIntegration.CheckFile
REBOOT	com.pax.poslink.PosLink
GETPINBLOCK	com.pax.poslink.fullIntegration.GetPinBlock

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RESETMSR	com.pax.poslink.PosLink
INPUTTEXT	com.pax.poslink.formManage.Input Text
AUTHORIZECARD	com.pax.poslink.fullIntegration.Authoriz eCard
COMPLETEONLINEEMV	com.pax.poslink.fullIntegration.CompleteOnlineEMV
REMOVECARD	com.pax.poslink.fullIntegration.RemoveCard
GETEMVTLVDATA	com.pax.poslink.fullIntegration.EMVTLVData
SETEMVTLVDATA	com.pax.poslink.fullIntegration.EMVTLVData
INPUTACCOUNTWITHEMV	com.pax.poslink.fullIntegration.InputAccount
SHOWTEXTBOX	com.pax.poslink.formManage.Input Text
CARDINSERTDETECTION	com.pax.poslink.PosLink
VASSETMERCHANTPARAMETERS	com.pax.poslink.fullIntegration.VasSetMerchantParameters
VASPUHDATA	com.pax.poslink.fullIntegration.VasPushData
SESSIONKEYINJECTION	com.pax.poslink.PosLink

7.2 Android BroadPOS P2PE/TermLink App External Calling (Need Install TermLink Manager Application)

7.2.1 External Calling Model



Picture 17 External Call Mode

7.2.2 UI Customization

In addition to the built-in UI, our application also supports customer-defined UI.

How to use:

You need enable the customUI support first.

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For Example:

```
1. CommSetting commSetting = new CommSetting();
2. commSetting.setCommType();
3. commSetting.setBaudRate();
4. commSetting.setPort();
5. commSetting.setCustomerUISupport(true);
6. ConnectManager.getInstance(this).startListeningService(commSetting, connectionCallBack);
```

Code 28 Example of ExternalCall CommonSetting

For the action command

Create an Activity. Set intent-filter, it contains action and category.

```
1. <activity
2.     android:name=".cusui.activity.EnterPinActivity"
3.     android:allowTaskReparenting="true"
4.     android:taskAffinity="com.pax.tlmanager">
5.     <intent-filter>
6.         <action android:name="com.pax.us.pay.action.ENTER_PIN" />
7.
8.         <category android:name="com.pax.us.pay.ui.category.SECURITY" />
9.         <category android:name="android.intent.category.DEFAULT" />
10.    </intent-filter>
11. </activity>
```

Code 29 Action and Category of CustomUI Activity

The following commands support custom UI with Activity:

Command	Intent-filter action	category
SHOWDIALOG	com.pax.us.pay.action.SHOW_DIALOG	com.pax.us.pay.ui.category.MANAGER
DOSIGNATURE	com.pax.us.pay.action.GET_SIGNATURE	com.pax.us.pay.ui.category.SIGNATURE
GETPINBLOCK	com.pax.us.pay.action.ENTER_PIN	com.pax.us.pay.ui.category.SECURITY
INPUTTEXT	com.pax.us.pay.action.INPUT_TEXT	com.pax.us.pay.ui.category.MANAGER
AUTHORIZECARD	com.pax.us.pay.action.ENTER_PIN	com.pax.us.pay.ui.category.SECURITY
INPUTACCOUNT-WITHEMV	com.pax.us.pay.action.INPUT_ACCOUNT	com.pax.us.pay.ui.category.SECURITY
	com.pax.us.pay.action.ENTER_EXPIRY_DATE	com.pax.us.pay.ui.category.TEXT
	com.pax.us.pay.action.ENTER_CVV	com.pax.us.pay.ui.category.SECURITY

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	com.pax.us.pay.action.ENTER_ZIP_CODE	com.pax.us.pay.ui.category.TEXT
	com.pax.us.pay.action.SELECT_AID	com.pax.us.pay.ui.category.OPTION
	com.pax.us.pay.action.SELECT_LANGUAGE	
SHOWTEXTBOX	com.pax.us.pay.action.SHOW_TEXT_BOX	com.pax.us.pay.ui.category.MANAGER
SHOWMESSAGECENTER	com.pax.us.pay.action.SHOW_MESSAGE_CENTER	com.pax.us.pay.ui.category.MANAGER

Let your Activity implements IUIListener interface and create a ActionHelper to process the message and callback. The ActionHelper will use IRespStatus to callback the transaction event and the user can use the function sendNext() and sendAbort() to go to the next step or cancel the transaction.

For Example:

```

1.  protected void onCreate(@Nullable Bundle savedInstanceState) {
2.      super.onCreate(savedInstanceState);
3.      mHelper = new EnterPinHelper(this, new IRespStatus() {
4.          @Override
5.          public void onAccepted() {
6.              //when trans accepted. Usually close the current activity
7.          }
8.          @Override
9.          public void onDeclined(long code, @Nullable String message) {
10.             //when trans declined with error message.
11.          }
12.          @Override
13.          public void onMessage(@Nullable String message) {
14.             //when trans need to send some message.
15.          }
16.          @Override
17.          public void onReset() {
18.             //when execute the reset command.
19.          }
20.          @Override
21.          public void onClearMsg() {
22.             //when execute the clearMessage command.
23.          }
24.      });
25.  }
26.  private void testOperate(){
27.      mHelper.sendNext();//goto the next step for this transaction.
28.      mHelper.sendAbort();//to cancel the current transaction.
29.  }
30.  @Override
31.  public void onShowAmount(long amount) {
32.      //set amount to textview or your label.
33.  }
34.  @Override
35.  public void onShowCurrency(@Nullable String currency, boolean isPoint) {
36.      //set currency symbol to textview or your label.
37.  }
38.  @Override
39.  public void onShowMessage(@Nullable String transName, @Nullable String message) {
40.      //set message to textview or your label.
41.  }
42.  @Override
43.  public void onShowPin(@Nullable String pinStyles, boolean isOnline, boolean isPinBypass) {
44.      //show your pin textbox or message.
45.  }
46.  @Override
47.  public void showDetail(String msg) {
48.      //you can show the detail message here.
49.  }

```

Code 30 Example of Callback for CustomUI Activity

The support ActionHelper for each action command like this:

Command	Action Helper	UI Listener	Recommand Activity Name
SHOWDIALOG	SelectOptionsHelper	ISelectOptionListener	ShowDialogActivity
DOSIGNATURE	SignatureHelper	ISignatureListener	DoSignatureActivity
GETPINBLOCK	EnterPinHelper	IEnterPinListener	EnterPINActivity
INPUTTEXT	InputTextHelper	IInputTextListener	InputTextActivity
AUTHORIZE CARD	EnterPinHelper	IEnterPinListener	EnterPINActivity
INPUTACCOUNTWITH V	SearCardHelper	ISearchCardListener	SearchCardActivity
	EnterExpiryDateHelper	IEnterExpiryDateListener	EnterExpiryDateActivity
	SecurityHelper	ISecurityListener	EnterCVVActivity
	EnterZipCodeHelper	IEnterZipCodeListener	EnterZipActivity
	SelectOptionsHelper	ISelectOptionListener	SelectOptionActivity
SHOWTEXTBOX	ShowTextBoxHelper	IShowTextBoxListener	ShowTextBoxActivity
SHOWMESSAGECENTER	ShowMessageHelper	IEnterAmountListener	ShowMessageCenterActivity

For the message command

The user can create a static BroadcastReceiver to receive the broadcast and set the intent-filter.

```

1. <receiver android:name=".cusui.receiver.StatusReceiver">
2.   <intent-filter>
3.     <action android:name="com.pax.us.pay.CARD_PROCESS_STARTED" />
4.     <action android:name="com.pax.us.pay.CARD_PROCESS_ERROR" />
5.     <action android:name="com.pax.us.pay.CARD_PROCESS_COMPLETED" />
6.
7.     <category android:name="com.pax.us.pay.status.category.CARD" />
8.   </intent-filter>
9.   <intent-filter>
10.    <action android:name="com.pax.us.pay.REMOVE_CARD_STARTED" />
11.    <action android:name="com.pax.us.pay.REMOVE_CARD_FINISHED" />
12.
13.    <category android:name="com.pax.us.pay.status.category.MANAGER" />
14.  </intent-filter>
15. </receiver>

```

Code 31 BroadcastReceiver for Transaction Status

```

1. <receiver android:name=".cusui.receiver.ClssStatusReceiver">
2.   <intent-filter>
3.     <action android:name="com.pax.us.pay.CLSS_LIGHT_NOT_READY" />
4.     <action android:name="com.pax.us.pay.CLSS_LIGHT_IDLE" />
5.     <action android:name="com.pax.us.pay.CLSS_LIGHT_READY_FOR_TXN" />
6.     <action android:name="com.pax.us.pay.CLSS_LIGHT_PROCESSING" />
7.     <action android:name="com.pax.us.pay.CLSS_LIGHT_REMOVE_CARD" />
8.     <action android:name="com.pax.us.pay.CLSS_LIGHT_COMPLETED" />
9.     <action android:name="com.pax.us.pay.CLSS_LIGHT_ERROR" />
10.    <action android:name="com.pax.us.pay.CLSS_LIGHT_BLUE_ON" />
11.    <action android:name="com.pax.us.pay.CLSS_LIGHT_BLUE_OFF" />
12.    <action android:name="com.pax.us.pay.CLSS_LIGHT_BLUE_BLINK" />
13.    <action android:name="com.pax.us.pay.CLSS_LIGHT_YELLOW_ON" />
14.    <action android:name="com.pax.us.pay.CLSS_LIGHT_YELLOW_OFF" />
15.    <action android:name="com.pax.us.pay.CLSS_LIGHT_YELLOW_BLINK" />
16.    <action android:name="com.pax.us.pay.CLSS_LIGHT_GREEN_ON" />
17.    <action android:name="com.pax.us.pay.CLSS_LIGHT_GREEN_OFF" />
18.    <action android:name="com.pax.us.pay.CLSS_LIGHT_GREEN_BLINK" />
19.    <action android:name="com.pax.us.pay.CLSS_LIGHT_RED_ON" />
20.    <action android:name="com.pax.us.pay.CLSS_LIGHT_RED_OFF" />
21.    <action android:name="com.pax.us.pay.CLSS_LIGHT_RED_BLINK" />
22.
23.   <category android:name="com.pax.us.pay.status.category.VIRTUAL_CLSS_LIGHT" />
24. </intent-filter>
25. </receiver>

```

Code 32 BroadcastReceiver for Contactless Light

```

1. <receiver android:name=".cusui.receiver.PinStatusReceiver">
2.   <intent-filter>
3.     <action android:name="com.pax.us.pay.PIN_CLEARED" />
4.     <action android:name="com.pax.us.pay.PIN_ENTER_COMPLETED" />
5.     <action android:name="com.pax.us.pay.PIN_ENTER_ABORTED" />
6.     <action android:name="com.pax.us.pay.PIN_ENTERING" />
7.     <action android:name="com.pax.us.pay.PIN_RNIB_TOUCH" />
8.     <action android:name="com.pax.us.pay.SECURITY_ENTERING" />
9.     <action android:name="com.pax.us.pay.SECURITY_DELETE" />
10.    <action android:name="com.pax.us.pay.SECURITY_FORMAT_ERROR" />
11.    <action android:name="com.pax.us.pay.SECURITY_LUHN_CHECK_ERROR" />
12.    <action android:name="com.pax.us.pay.SECURITY_LUHN_CHECK_OK" />
13.
14.    <category android:name="com.pax.us.pay.status.category.PIN" />
15.    <category android:name="com.pax.us.pay.ui.category.SECURITY" />
16.  </intent-filter>
17. </receiver>

```

Code 33 BroadcastReceiver for Security Input status

The user can use the EventBus or other ways to notify your activity to display a dialog box or process related content.

For Example:

In BroadcastReceiver

```

1. public class PinStatusReceiver extends android.content.BroadcastReceiver {
2.
3.   @Override
4.   public void onReceive(Context context, Intent intent) {
5.     Log.d("PinStatusReceiver", "onReceive: " + intent.getAction());
6.     if (TextUtils.isEmpty(intent.getAction()))
7.       return;
8.     EventBus.getDefault().post(intent.getAction());
9.   }
10. }

```

Code 34 BroadcastReceiver code example

In Activity

```

1.  @Override
2.  protected void onCreate(@Nullable Bundle savedInstanceState) {
3.      super.onCreate();
4.      EvensBus.getDefault().register(this);
5.  }
6.
7.  @Override
8.  protected void onDestroy() {
9.      EventBus.getDefault().unregister(this);
10.     super.onDestroy();
11. }
12.
13. @Subscribe(threadMode = ThreadMode.MAIN)
14. public void onEventAsync(String action) {
15.     switch (action) {
16.         case PINStatus.PIN_ENTERING:
17.             Log.i("EnterPinEvent", "PINStatus.PIN_ENTERING");
18.             pinLen++;
19.             String star = "*";
20.             StringBuilder sb = new StringBuilder();
21.             for (int i = 0; i < pinLen; i++) {
22.                 sb.append(star);
23.             }
24.             pwdInputText.setText(sb.toString());
25.             break;
26.         case PINStatus.PIN_ENTER_CLEARED:
27.             Log.i("EnterPinEvent", "PINStatus.PIN_ENTER_CLEARED");
28.             pinLen = 0;
29.             pwdInputText.setText("");
30.             break;
31.         case PINStatus.PIN_RNIB_TOUCH:
32.             Log.i("EnterPinEvent", "PINStatus.PIN_RNIB_TOUCH");
33.             break;
34.     }
35. }

```

Code 35 Example of Security Input Status usage

The following commands or messages support custom UI with BroadcastReceiver:

Command or Message	Intent-filter action	category
REMOVECARD	com.pax.us.pay.REMOVE_CARD_STARTED com.pax.us.pay.REMOVE_CARD_FINISHED	com.pax.us.pay.status.category.MANAGER
CARD STATUS	com.pax.us.pay.CARD_PROCESS_STARTED com.pax.us.pay.CARD_PROCESS_COMPLETED	com.pax.us.pay.status.category.CARD
CLSS LIGHT	com.pax.us.pay.CLSS_LIGHT_NOT_READY com.pax.us.pay.CLSS_LIGHT_READY_FOR_TXN	com.pax.us.pay.status.category.VIRTUAL_CLSS_LIGHT

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	com.pax.us.pay.CLSS_LIGHT_PROCESSING com.pax.us.pay.CLSS_LIGHT_REMOVE_CARD com.pax.us.pay.CLSS_LIGHT_COMPLETED com.pax.us.pay.CLSS_LIGHT_ERROR com.pax.us.pay.CLSS_LIGHT_BLUE_ON com.pax.us.pay.CLSS_LIGHT_BLUE_OFF com.pax.us.pay.CLSS_LIGHT_BLUE_BLINK com.pax.us.pay.CLSS_LIGHT_YELLOW_ON com.pax.us.pay.CLSS_LIGHT_YELLOW_OFF com.pax.us.pay.CLSS_LIGHT_YELLOW_BLINK com.pax.us.pay.CLSS_LIGHT_GREEN_ON com.pax.us.pay.CLSS_LIGHT_GREEN_OFF com.pax.us.pay.CLSS_LIGHT_GREEN_BLINK com.pax.us.pay.CLSS_LIGHT_RED_ON com.pax.us.pay.CLSS_LIGHT_RED_OFF com.pax.us.pay.CLSS_LIGHT_RED_BLINK	
PIN SATAUS	com.pax.us.pay.PIN_CLEARED com.pax.us.pay.PIN_ENTER_COMPLETED com.pax.us.pay.PIN_ENTER_ABORTED com.pax.us.pay.PIN_ENTERING com.pax.us.pay.PIN_RNIB_TOUCH com.pax.us.pay.SECURITY_ENTERING com.pax.us.pay.SECURITY_DELETE	com.pax.us.pay.status.category.PIN com.pax.us.pay.ui.category.SECURITY

8 Appendix

8.1 Classic Terminal EMV Kernel Version

Application Provider	PAX TECHNOLOGY, INC.
Address	4/F., No.3 Building, Software Park, 2nd Central Science-Tech Road, High-Tech Industrial Park Guangdong Shenzhen 518057 CHINA
Payment model:	S300, Q20, Q25, Q30, PX5, IM20, IM300, IM500, IM700, PX7, D190
Product Description:	POS Terminal
Product Options:	S300, Q20, Q25, Q30, PX5, IM20, IM300, IM500, IM700, PX7, D190

Run below contactless kernel in PAX BroadPOS P2PE Applications, version 1.xx.xx, and PAX TermLink Application, version 1.xx.xx

- F WAVE LIB v353
- F MC LIB v552_01
- F DPAS LIB v151_1
- F AE LIB v250_02
- F QPBOC v351
- F JCB LIB v151_1
- F FLASH LIB v250

Run below contact kernel in PAX BroadPOS P2PE Applications, version 1.xx.xx, and PAX TermLink Application, version 1.xx.xx

- EMV_LIB_v653

8.2 Android Terminal EMV Kernel Version

Application Provider	PAX TECHNOLOGY, INC.
Address	4/F., No.3 Building, Software Park, 2nd Central Science-Tech Road, High-Tech Industrial Park Guangdong Shenzhen 518057 CHINA
Payment model:	A30, A35, A50, A60, A77, A80, A920, A920Pro, A930, AR6, AR8, IM30, PX7A
Product Description:	POS Terminal
Product Options:	A30, A35, A50, A60, A77, A80, A920, A920Pro, A930, AR6, AR8, IM30, PX7A

Run below contactless kernel in PAX BroadPOS P2PE Applications, version 1.xx.xx, and PAX TermLink Application, version 1.xx.xx

- F WAVE LIB v353
- F MC LIB v553
- F DPAS LIB v152

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- F AE LIB v250_03
- F QPBOC v352
- F JCB LIB v150_01
- F FLASH LIB v250

Run below contact kernel in PAX BroadPOS P2PE Applications, version 1.xx.xx, and PAX TermLink Application, version 1.xx.xx

- EMV_LIB_v653

8.3 Offline Data Verification For Contactless

PayPass:

Only support the CDA, the <SecurityCapability> bit 4 can set the CDA.

PayWave:

Only support the fDDA, the specification requires that fDDA must be turned on for terminals that support offline, it cannot be turned off by APP now.

ExpressPay:

Support SDA and CDA, the <TerminalCapability> B3b4 to set CDA, and the B3b8 to set SDA. the specification requires that CDA and SDA must be turned on for terminals, the risk of closing SDA and CDA is borne by the customer.

D-PAS:

Only support the CDA, the specification requires that CDA must be turned on for terminals, it cannot be turned off.

JSpeedy:

Legacy Mode and Magstripe Mode does not support CDA, only EMV mode support CDA. The <CombinationOptions> B1b6 to set CDA, the specification requires that CDA must be turned on for terminals, the risk of closing SDA and CDA is borne by the customer.

QPBOC:

Only support the fDDA, the specification requires that fDDA must be turned on for terminals that support offline, it cannot be turned off by APP now.

8.4 Control Character for Text Showing

Character	Description	Commands
\n	Line feed	ShowMessage ShowMessageCenter ShowTextBox
\C	Center alignment	ShowMessage ShowMessageCenter ShowTextBox
\L	Left alignment	ShowMessage ShowMessageCenter ShowTextBox
\R	Right alignment	ShowMessage ShowMessageCenter ShowTextBox
\B	Bold word	ShowMessage ShowMessageCenter ShowTextBox

Notes:

1. “\C” “\L” “\R” will not change line automatically, please add “\n” if you want to change to next line.
2. “\B” can be used along with “\L” or “\R” or “\C” with format like “\B\R”.
3. “\B” will automatically jump to next line if there are some words before “\B”.
4. “\B” will only bold the first line.

Control Character	Description	command
\1	Small font	ShowMessage ShowMessageCenter ShowTextBox
\2	Normal font(default)	ShowMessage ShowMessageCenter ShowTextBox
\3	Big font	ShowMessage ShowMessageCenter ShowTextBox

8.5 Terminal Supported By BroadPOS P2PE/TermLink

Application

8.5.1 Android Terminal

Terminal	Release Date
A920	2019/6
A60	2019/7
A80	2019/7
A930	2019/7
Aries6	2019/9

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Aries8	2019/9
PX7A	2019/9
IM30	2019/11
A50	2020/6
A30	2020/9
A77	2020/9
A920Pro	2020/9
A60	2020/10
A35	2021/8

8.5.2 Classic Terminal

Terminal	Release Date
S300	2019/8
Q30	2019/10
PX5	2019/10
PX7	2019/10
IM300	2019/12
IM20	2020/1
Q20	2020/9
D190	2020/10
IM500	2021/5
IM700	2021/5